The Incidence of Carbon Taxes in an Electrifying Economy

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Abstract

When household electrification is taken into account, the carbon tax incidence becomes age-specific, falling hardest on middle-aged, low-income households. It also creates an equity–efficiency trade-off in how carbon tax revenues are recycled. I develop a life cycle model with a discrete household electrification decision, and include energy as a necessity good and production input. The government taxes carbon to achieve a climate target, and recycles revenues via government spending, lump-sum rebates, renewable-electricity subsidies, or electrification subsidies. Using renewable-electricity subsidies reduces the required carbon tax from €359/tCO_2 to €248/tCO_2, cutting wage declines by half. However, negative welfare effects concentrate among middle-aged, low-income households, for whom electrifying is as costly as remaining exposed to the tax. The electrification margin explains 43% of aggregate welfare losses, which are concentrated among middle-aged, low-income households. Lump-sum transfers insure these households against the burden from electrification, albeit with larger wage declines.

JEL classification: D31, H22, H23, Q48, Q54

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1 Introduction

How does household electrification change who bears the costs of a carbon tax? Households differ in their exposure to carbon pricing because they consume different fuels: electrified households consume electricity, others remain directly exposed to taxed fossil fuels. This makes electrification a discrete adaptation margin that determines exposure to the carbon tax. In this paper, I analyze how the discrete decision to electrify shapes the incidence of a carbon tax, and how this incidence depends on the way tax revenues are recycled. For households, electrifying means replacing a gas boiler and a petrol car with a heat pump and an electric vehicle. This investment requires a large upfront payment, often unaffordable for liquidity-constrained poor households (Berg et al., 2024), and pays off only gradually through lower future energy bills. The electrification decision therefore depends not only on income and wealth, but also on age: younger households have more time to recover the investment than older households. Households choose between paying the upfront cost of electrification and benefiting from lower future energy expenditures, or remaining dependent on fossil fuels and paying the carbon tax in full over their remaining lifetime. Recycling revenues through green electricity subsidies can lower the carbon tax required to achieve a given climate target and thereby reduce aggregate distortions, but benefits accrue disproportionately to households that electrify early. As a result, the policy that makes the transition cheapest for the economy need not protect the households that bear its largest costs.

This paper makes three contributions. First, I show that an electrification margin makes the welfare incidence of a carbon tax vary sharply with age on top of income, and fall hardest on middle-aged, low-income households. Since households recover the upfront cost of electrification only later, through lower energy bills, the remaining lifetime horizon matters for the electrification decision. While the electrification margin accounts for 43% of the aggregate welfare loss, it explains up to 90% of it among middle-aged, low-income households. The combination of a life cycle and a discrete electrification margin concentrates the welfare-loss at the middle-aged, low-income households. This age-incidence matters for the politics of carbon taxes too: households between 40 and 65 are among the strongest opponents of the policy (Levi, 2021).

The main mechanism in the model that creates this hump-shaped tax incidence comes from the fact that for middle-aged, low-income households switching to a green durable is as costly as paying the carbon tax for their remaining lifetime. Households who do not electrify are directly exposed to the carbon tax, and because energy is a necessity, their ability to reduce energy demand is limited. On the other hand, households who do electrify pay a large upfront cost. The large upfront payment is concentrated and near-term, while pay-offs only materialize via lower energy expenditures in the future. Thus while the costs of electrifying, i.e. user-costs of the durable good, increase in age, the costs of not electrifying, i.e. the remaining energy expenditures, decrease in age. The age group with the largest increase in expenditures (either on energy, or green durable goods) following the carbon tax is at the age of intersection of these costs.

Second, I characterize an equity–efficiency trade-off in the choice of revenue recycling instrument (subsidies versus transfers) which arises when household electrification and electricity sector decarbonization are modeled jointly. The supply side determines the *size* of both the distortions the carbon tax causes and the gains a subsidy can deliver; the electrification margin largely determines the *incidence* of these losses and gains. Green subsidies lower the peak carbon tax required for an exogenous target, and thus reduce aggregate GDP and wage distortions, but because richer households electrify earlier, the subsidies accrue mostly to them via lower electricity prices or lower electrification costs. Shifting revenue toward lump-sum redistribution lowers long-run GDP, but improves welfare for low-income households, who receive a larger share of the tax revenue directly. Models that assess the detailed effects of either margin alone do not capture this trade-off (Ascari et al., 2025; Hochmuth et al., 2025; Kuhn & Schlattmann, 2024; Crucitti et al., 2024).

The model makes the trade-off quantitative. Households differ in income (à la Aiyagari (1994)), wealth, age and electrification status. Electrification is a discrete, forward-looking decision modeled as a state variable: a household with a brown durable can pay an upfront cost, potentially using a collateralized loan, to switch to a green durable. The type of durable determines the energy source (fossil fuels for the brown, electricity for the green) and converts it into services of mobility and heating. A representative final-goods firm combines capital, labor, and an energy composite of electricity and imported fossil fuels; the electricity sector aggregates renewable and fossil generated electricity. Both the final-goods producer and the electricity sector pay the carbon tax on their fossil-fuel inputs.

I examine three ways in which the government can ease the burden of a carbon tax using revenue recycling instruments, relative to wasteful government spending: (1) lump-sum transfers, (2) renewable electricity subsidies, and (3) electrification subsidies. For each instrument, I solve for the carbon tax that achieves a target temperature path and abstract from climate damages so that the instruments are compared internally on equal terms. The efficiency case for the subsidy is real, but narrow and only holds when subsidies are targeted at renewable electricity production. Subsidizing renewable electricity lowers the required carbon tax and electricity price, so that long-run GDP falls only half as much as under lump-sum recycling. An electrification subsidy produces no comparable gain, since it barely affects energy prices. On equity, electricity and electrification subsidies impose the largest welfare losses on middle-aged, low-income households, almost entirely through the electrification margin following the above mechanism. Lump-sum redistribution fully offsets this burden by offering broad consumption insurance to these households.

Finally, I use the model to study the timing of carbon taxation. Front-loading curbs emissions sooner, so a lower peak tax rate reaches the same climate target. However, I show that the welfare effect of front-loading changes sign with the recycling instrument. A faster implementation of the carbon tax combined with green subsidies amplifies welfare losses in the presence of the electrification margin, especially for the poor who cannot adapt. This is caused by a higher carbon tax in the short-run, as a consequence of front-loading. Pairing front-loading with lump-sum redistribution reverses this: transfers are higher in the short run, raising welfare for low-income households.

The remainder of this paper is structured as follows. I relate this paper to the existing literature in section 1.1. In section 2, I discuss the model that I parametrize in section 3. In section 4, I conduct policy experiments and analyze the results for equity and efficiency. I assess robustness in section 5. Lastly, I analyze how the results change when I front-load the tax in section 6, and conclude in section 7.

1.1 Related literature

A growing quantitative literature studies inequality along the green transition. A first strand treats energy purely as an input into production affecting inequality through labor income and revenue redistribution (Benmir & Roman, 2022; Boehl et al., 2024). Another strand treats energy as both a production input and something households consume directly. Känzig (2023) models energy as a fossil fuel that enters both production and a non-homothetic utility function. A carbon tax affects households twice: indirectly, via lower income, and directly, via their own energy use. Fried et al. (2024) use a similar non-homothetic setup to compare carbon-tax rebate instruments, and find that cutting distortionary taxes and raising the progressivity of labor taxes raises welfare more than lump-sum transfers do. I instead allow households to differ endogenously in the fuel they consume through a discrete electrification decision.

Other papers make this differentiation too, so that climate policy can cut emissions by decarbonizing electricity itself rather than only by reducing its consumption. Ascari et al. (2025) model two energy sources, fossil and renewable, that a profit-maximizing firm combines without friction into a single bundle sold to every household and firm; a carbon tax raises the fossil price and shifts the cost-minimizing mix toward renewables. Because every household consumes the same bundle, inequality in their model comes entirely from non-homotheticity, energy's share of spending falls as income rises. Hochmuth et al. (2025) model the same structure and show that the elasticity of substitution between green and brown energy, together with the degree of non-homotheticity, set the size of the inequality effect. In both papers, every household consumes the same energy mix. In my paper, fuel type consumption is heterogeneous between households by giving households an electrification choice. Consequently, a household's exposure depends on whether it has electrified or not, on top of its income via non-homothetic preferences. The electricity good in my model follows the same structure as above: a CES aggregate of renewable and fossil generation.

Labrousse & Perdereau (2024) and Schlattmann (2024) add this heterogeneous fuel consumption via a spatial margin: households choose where to live, and rural households, who rely more on fossil energy, bear a larger burden from a carbon tax. Schlattmann (2024) also lets households choose between a green and a brown car. Where their margin is location, mine is the upfront cost of electrifying in place.

This electrification margin in itself is studied in Kuhn & Schlattmann (2024). Using a partial equilibrium model, they assess how a subsidy for switching from an old, more polluting durable to a modern one affects inequality and transition speed under different financing schemes. Closely related, Crucitti et al. (2024) study heat pump adoption over the life cycle under a similar upfront cost, and find that a proportional subsidy benefits older households more than a lump-sum one does, because they own larger homes; Schlattmann (2024), discussed above, finds the same pattern for house size. I set aside these size effects and instead focus only on the discrete decision to switch technology. Unlike in Kuhn & Schlattmann (2024) and Crucitti et al. (2024) I let the electricity sector itself decarbonize. Letting electricity supply respond too is what separates the size of the gains a subsidy creates from who actually receives them. Lastly, Fried (2026) studies a related adoption margin driven by climate damages rather than climate policy: richer households adopt heat pumps to manage temperature extremes. In that paper, temperature variance drives the adoption of heat pumps.

Finally, Fried et al. (2018) compare steady states to study how the burden of a carbon tax falls across generations under different recycling rules. I instead follow households through the transition itself to assess the effect of a carbon tax on different generations.

2 Model

Overlapping generations of households populate the economy. During its working life the household faces idiosyncratic productivity shocks. On top of a standard consumption-savings model, I add two features that allow me to analyze the effect of an electrification margin.

First, I add energy consumption by the household and include a subsistence requirement to capture the non-homothetic preference for energy services. I differentiate between two sources of energy, fossil fuels and electricity. The second addition is a durable good that governs the type of energy the household consumes. A household owns either a brown durable and consumes fossil fuels, or a green durable and consumes electricity. When switching from a brown to a green durable, hence electrifying, the household can use a collateralized loan. The durable good and loan are modeled as state variables. Throughout, all state variables chosen in the current period and carried into the next are denoted by a prime.

On the production side there are three sectors: a final-consumption good sector, an electricity sector, and a durable goods sector. Lastly, a government collects carbon tax revenues and recycles them within the same period.

2.1 Households

Demographics Time is discrete. Households' age is indexed by $j = 0, 1, 2, ..., J-1$. In period J the household dies with certainty. A household works for J_w periods, and then retires. The economy is populated by a continuum of households with mass one. At birth, households differ in their income $y_{i,0}$, initial wealth $a_{i,0}$ and their durable good, $\phi_{i,0}$, which is a binary variable indicating whether the household owns a brown or green durable. Households are indexed by i, however for notational convenience this index is dropped in most cases.

Preferences Households maximize expected lifetime utility

$$U = E_0\left[\sum_{j=0}^{J-1} \beta^j u(c_j, \mathcal{S}(\phi_j, e_j)) + \beta^J \mathcal{W}(b)\right], \tag{1}$$

where β is the discount factor, c_j is consumption of the final consumption good, $\mathcal{S}(\phi_j, e_j)$ the flow of energy services, and $\mathcal{W}(b)$ is a warm-glow bequest. Flow utility is a Stone-Geary utility function nested in a CRRA

$$u(c, \mathcal{S}(\phi, e)) = \frac{1}{1-\sigma}(c^\gamma(\mathcal{S}(\phi, e) - \bar{\mathcal{S}})^{1-\gamma})^{1-\sigma}, \tag{2}$$

where γ denotes the households' relative taste for the final consumption good, $\frac{1}{\sigma}$ is the intertemporal elasticity of substitution, and $\bar{\mathcal{S}}$ is a subsistence level for services derived from energy. The subsistence level captures the non-homotheticity in energy demand.[1]

[1]Most papers either use a subsistence level to capture this non-homotheticity Wöhrmüller (2024); Schlattmann (2024); Fried et al. (2024) or the flexible indirect utility function of Boppart (2014), adopted by among others Ascari et al. (2025); Hochmuth et al. (2025). Formally, the former is quasi-homothetic, as

The amount of energy services, $\mathcal{S}(\phi, e)$, depends on the level of energy consumption, e, and the type of durable good, ϕ. A household owns a green durable, d_G, or a brown durable, d_B. For notational convenience I denote this by a binary variable, $\phi \in \{0, 1\}$, such that $d_\phi \equiv \phi d_G + (1-\phi) d_B$. The type of the durable good, ϕ, sets the rate at which energy, e, translates into services, such that

$$\mathcal{S}(\phi, e) = \begin{cases} A_{s,G}e & : \text{if } \phi = 1 \implies d_\phi = d_G \\ A_{s,B}e & : \text{if } \phi = 0 \implies d_\phi = d_B. \end{cases} \tag{3}$$

Akin to the technologies in Fried (2026), each durable good transforms energy (here fossil fuels or electricity) into services of mobility and heating at a different rate. A higher rate implies a more energy efficient durable good. The energy efficiency of the durable good is thus pinned down by its productivity, $A_{s,G}$ for the green and $A_{s,B}$ for the brown durable good.[2]

A dying household receives utility from bequeathing its savings and durable good as

$$\mathcal{W}(b) = \eta \frac{(b + \underline{b})^{1-\sigma}}{1-\sigma}, \tag{4}$$

with bequest strength η, and where $\underline{b}$ denotes the extent to which bequeathing is a luxury good, as in De Nardi (2004). The size of the bequest is given by

$$b_t = a'_{J-1,t} + ((1 - \phi'_{J-1,t})P^{d_B} + \phi'_{J-1,t} P_t^{d_G}), \tag{5}$$

where P^{d_ϕ} denotes the price of the durable good.

Durable good and energy As explained above, the economy has two types of durable goods: a green durable d_G and a brown durable d_B, and converts energy into services (mobility, heating) at its own rate, $A_{s,G}$ or $A_{s,B}$. Furthermore, the green durable runs on electricity, while the brown durable requires fossil fuels. Switching from a brown to a green durable is therefore called electrification.

A newborn household receives either type of durable good via a warm-glow bequest. The probability of receiving the green durable depends on the household's initial income

it gives linear marginal budget shares in the interior (away from the subsistence level) whereas the latter is fully non-homothetic. Quasi-homotheticity is sufficient to capture that poor households cannot substitute away from energy when the price increases.

[2]These productivities are calibrated as the coefficient of performance, as defined by the International Energy Agency of the respective durable goods. The rate at which fossil fuels (or electricity) being translated into services of mobility and heating by an electric vehicle (heat pump) and gasoline vehicle (regular boiler).

level and follows the dying cohorts' joint distribution of income and durables. A household electrifies by selling its brown durable for a scrap value χP^{d_B} and buying a green durable for $P_t^{d_G}(1 - s_{d_G,t})$, where $s_{d_G,t}$ is a subsidy for electrifying households. Electrification is an absorbing state: households cannot switch back from green to brown. When a household electrifies, it can use a secured borrowing scheme m, subject to a down-payment constraint

$$m' \leq \lambda^{LTV} P_t^{d_G}(1 - s_{d_G,t}). \tag{6}$$

The household pays an interest rate premium r_m on the loan and repays following an amortization schedule, so that the fraction of the loan due each period is

$$\pi(j,t) = \frac{(r_t + r_m)(1 + (r_t + r_m))^{J-j}}{(1 + (r_t + r_m))^{J-j} - 1}, \tag{7}$$

giving next period's loan $m' = (1 + r_t + r_m - \pi(j,t))m$. Each durable good requires maintenance costs equal to the depreciation rate $\delta_\phi = \phi\delta_{d_G} + (1-\phi)\delta_{d_B}$, paid every period.

The fuel price per unit of energy that the household faces depends on the durable good it owns. As noted, the green durable good relies on electricity, E with price P_t^E, whereas the brown durable good relies on fossil fuels, FF, with price subject to a carbon tax, $P^{FF} + \tau_{c,t}$.

Income and wealth Households supply their labor inelastically and receive income as an endowment each period. Labor income is denoted as $w_t y_{i,j}$, where $y_{i,j}$ denotes labor productivity of a household i at age j, and is given by

$$log(y_{i,j}) = \xi(j) + z_{i,j}. \tag{8}$$

such that labor productivity $y_{i,j}$ consists of a deterministic lifecycle component $\xi(j)$ and an individual component, $z_{i,j}$, representing productivity differences between households. During the first J_w years, the idiosyncratic part of labor productivity $z_{i,j}$ is stochastic and follows an AR(1) process,

$$z_{i,j} = \rho_z z_{i,j} + \epsilon_{i,j}. \tag{9}$$

After J_w periods, the household retires. It then deterministically receives a fixed proportion f_{ret} of its last-period income as a pension for the remaining $J - J_w$ periods.

Households save in a risk-free asset, a, earning interest rate, r_t, every period. Unsecured borrowing is not allowed. Initial wealth is zero with probability p_0. If initial wealth is non-zero, it is drawn from a univariate log-normal wealth distribution with variance $\sigma_{a_0}^2$, which integrates up to the total bequests from the dying households in that period.

2.1.1 Recursive formulation household problem

A household is born with initial wealth a_0, income y_0, and durable good ϕ_0 (for all households $m_0 = 0$). Thereafter, the household solves the following problem

$$V_j(a, \phi, m, y; P_t, \Omega_t) = \max\{V_j^{NA}(a, \phi, m, y; P_t, \Omega_t), V_j^{Electrify}(a, \phi, m, y; P_t, \Omega_t) + \mathcal{E}_{adj}\} \quad (10)$$

where $P_t : \{r_t, w_t, P_t^E, P_t^{FF}, P_t^{d_G}, P_t^{d_B}\}$ denotes the price vector and $\Omega_t : \{\tau_{c,t}, \mathcal{T}_t, s_{E_R,t}, s_{d_G,t}\}$ the vector of government policies. Here $\mathcal{T}_t$ denotes a lump-sum transfer, and the carbon tax and subsidies are defined as above. I drop the time subscript for individual state and choice variables for readability. I introduce $\mathcal{E}_{adj}$ as an iid adjustment shifter following a Gumbel distribution with location parameter $\mu_{adj} < 0$ and scale parameter σ_{adj}. The shifter induces a reduced-form probabilistic adjustment rule that smoothly allocates the mass of otherwise identical households across the two choices. The probability of electrifying is an increasing function in the value difference $V_j^{\text{Electrify}}(\cdot) - V_j^{NA}(\cdot)$.[3] A non-electrifier solves

$$V_j^{NA}(a, \phi, m, y; P_t, \Omega_t) = \max_{a', c, e} u(c, \mathcal{S}(\phi, e)) + \beta E[V_{j+1}(a', \phi', m', y'; P_{t+1}, \Omega_{t+1})], \quad (11)$$

subject to

$$w_t y + \mathcal{T}_t + (1 + r_t)a \geq c + a' + e[(1 - \phi)(P^{FF} + \tau_{c,t}) + \phi P_t^E] + \delta_\phi P_t^{d_\phi} + \pi(j, t)m, \quad (12)$$

$$a' \geq 0, \quad (13)$$

$$m' = (1 + (r_t + r_m) - \pi(j, t))m, \quad \phi' = \phi. \quad (14)$$

Equation 12 is the budget constraint: labor income, a lump-sum transfer and asset income must cover consumption, savings, energy expenditures, depreciation and repayments on any outstanding loan. The remaining equations give the unsecured borrowing constraint and the evolution of the loan repayment.

[3] The adjustment shifter is introduced primarily as a numerical smoothing device. In the numerical implementation, I construct adjustment probabilities and propagate ex-ante continuation values as probability-weighted averages, i.e. the continuation value is then $p^{adj} V^{\text{Electrify}} + (1 - p^{adj}) V^{NA}$, where p^{adj} denotes the probability of electrification implied by the Gumbel specification. When computing marginal continuation values in the EGM step, these probabilities are held fixed. Details on the calculation of p^{adj} are provided in Online Appendix A. The adjustment shifter affects household behavior through the probability of electrification (an iid dislike to change) but is not treated as a welfare-relevant component of preferences. Welfare is evaluated from consumption, energy services, and bequests generated by the resulting allocation.

The value function for a household who chooses to electrify is

$$V_j^{Electrify}(a, \phi, m, y; P_t, \Omega_t) = \max_{a',m',c,e} u(c, \mathcal{S}(\phi, e)) + \beta E[V_{j+1}(a', \phi', m', y'; P_{t+1}, \Omega_{t+1})], \quad (15)$$

subject to

$$w_t y + \mathcal{T}_t + (1+r_t)a + m' \geq c + a' + e\phi' P_t^E + \delta'_\phi P_t^{d'_\phi} + (P_t^{d_G}(1 - s_{d_G,t}) - \chi P^{d_B}), \quad (16)$$

$$a' \geq 0, \quad (17)$$

$$m' \leq \lambda^{LTV} P_t^{d_G}(1 - s_{d_G,t}), \quad \phi' = 1. \quad (18)$$

Again, equation 16 is the budget constraint: all income, consisting of labor, a lump-sum transfer and wealth income, plus the loan taken out should exceed consumption, savings, energy expenditure, depreciation costs and the purchasing cost of the green durable minus the resale value of the brown.

Lastly $V_J = \mathcal{W}(b)$ is as in equation 4.

2.2 Production

The production side of the economy has three sectors: a final consumption good sector producing Y^C, an electricity sector producing Y^E and, a durable goods sector producing the green and brown durables Y^{d_G} and Y^{d_B}. The economy imports fossil fuels.

Final consumption good A representative firm produces a final good Y^C using capital K_C labor L_C, electricity E_C, and fossil fuels FF_C. The good serves as either a consumption good or an input into durable good production. The production technology is a constant-returns-to-scale nested CES:

$$Y^C(K_C, L_C, E_C, FF_C) = \left[(1-\psi)^{\frac{1}{\nu}} \left(A_{KC} K_C^{\alpha_c} L_C^{1-\alpha_c}\right)^{\frac{\nu-1}{\nu}} + \psi^{\frac{1}{\nu}} \left(M(E_C, FF_C)\right)^{\frac{\nu-1}{\nu}}\right]^{\frac{\nu}{\nu-1}}, \quad (19)$$

where $\psi \in (0,1)$ is the share of the energy composite, $\nu > 0$ the elasticity of substitution between the capital-labor bundle and the energy composite, $\alpha_c \in (0,1)$ the capital share, and A_{KC} capital-labor bundle productivity. The energy composite $M(E_C, FF_C)$ nests electricity and fossil fuels:

$$M(E_C, FF_C) = \left[\kappa^{\frac{1}{\iota}} (A_{EC} E_C)^{\frac{\iota-1}{\iota}} + (1-\kappa)^{\frac{1}{\iota}} (A_{FFC} FF_C)^{\frac{\iota-1}{\iota}}\right]^{\frac{\iota}{\iota-1}}. \quad (20)$$

where $\kappa \in (0,1)$ is the electricity share in the energy composite, $\iota > 0$ the elasticity of substitution between electricity and fossil fuels, and A_{EC} and A_{FFC} the respective productivities. The firm takes the electricity price, P_t^E, the carbon-tax-included fossil fuel price, $(P^{FF} + \tau_{c,t})$, the wage, w_t, and the rental cost of capital, $(r_t + \delta_K)$ as given, and maximizes profits:

$$\Pi^C = Y^C - w_t L_C - (r_t + \delta_K) K_C - P_t^E E_C - (P^{FF} + \tau_{c,t}) FF_C. \tag{21}$$

Electricity A representative firm produces electricity by aggregating renewable electricity Y^{E_R} and fossil-generated electricity Y^{E_F}. I use a similar technology to that in Ascari et al. (2025), however, given the capital intensity of the sector, I simplify by assuming labor is not a production input. Renewable electricity uses capital alone:

$$Y^{E_R}(K_{E_R}) = A_{KR,t} K_{E_R}, \tag{22}$$

K_{E_R} is capital allocated to renewable generation and $A_{KR,t}$ its productivity. The time subscript denotes that this productivity increases over time to capture an exogenous part of the energy transition. Fossil electricity combines capital K_{E_F} and fossil fuels FF_E via a CES technology:

$$Y^{E_F}(K_{E_F}, FF_E) = \left[\zeta^{\frac{1}{\omega}} (A_{KF} K_{E_F})^{\frac{\omega-1}{\omega}} + (1-\zeta)^{\frac{1}{\omega}} FF_E^{\frac{\omega-1}{\omega}} \right]^{\frac{\omega}{\omega-1}}, \tag{23}$$

where $\zeta \in (0,1)$ is the capital share, $\omega > 0$ the elasticity of substitution between capital and fossil fuels, and A_{KF} capital productivity in fossil generation. The firm aggregates these into total electricity output:

$$Y^E(K_{E_R}, K_{E_F}, FF_E) = \left[\alpha_R^{\frac{1}{\lambda}} \left(Y^{E_R}\right)^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} \left(Y^{E_F}\right)^{\frac{\lambda-1}{\lambda}} \right]^{\frac{\lambda}{\lambda-1}}, \tag{24}$$

where $\alpha_R \in (0,1)$ and $\alpha_F = 1 - \alpha_R$ are the shares of renewable and fossil electricity, and $\lambda > 0$ the elasticity of substitution between them. The firm takes the electricity price P_t^E, the carbon-tax-included fossil fuel price $(P^{FF} + \tau_{c,t})$, and the rental cost of capital $(r_t + \delta_K)$ as given, and maximizes profits:

$$\Pi^E = P_t^E Y^E - (r_t + \delta_K)(1 - s_{E_R,t}) K_{E_R} - (r_t + \delta_K) K_{E_F} - (P^{FF} + \tau_{c,t}) FF_E, \tag{25}$$

where $s_{E_R,t}$ is a government subsidy on the capital cost of renewable generation.

Durable goods Two representative firms produce the green and brown durable goods by converting the final consumption good (the numeraire, with price normalized to one) at conversion rates $A_{G,t}$ and A_B:

$$Y^{d_G}(C_G) = A_{G,t}\, C_G, \tag{26}$$
$$Y^{d_B}(C_B) = A_B\, C_B, \tag{27}$$

where C_G and C_B are the quantities of the consumption good used as inputs. The time subscript on $A_{G,t}$ denotes that green durable productivity increases over time, capturing cost reductions in the green technology. Each firm maximizes profits:

$$\Pi^{d_G} = P_t^{d_G}\, Y^{d_G} - C_G, \tag{28}$$
$$\Pi^{d_B} = P^{d_B}\, Y^{d_B} - C_B. \tag{29}$$

In equilibrium, profits are zero, which pins down durable good prices: $P_t^{d_G} = 1/A_{G,t}$ and $P^{d_B} = 1/A_B$.

2.3 Government

The government runs a balanced budget. It receives carbon tax revenues, and spends it on government expenditure, a lump-sum transfer, a subsidy on the costs to generate renewable electricity generation or a subsidy for the costs of electrification, such that every period

$$\begin{aligned} G_t + \mathcal{T}_t + s_{E_R,t} K_{E_R,t}(r_t + \delta_K) + s_{d_G,t} P_t^{d_G} \sum_{j=0}^{J-1} \int (\phi' - \phi) d\mu_{j,t} \\ = \tau_{c,t}[FF_{HH,t} + FF_{E,t} + FF_{C,t}]. \end{aligned} \tag{30}$$

2.4 Equilibrium

Given an initial distribution over initial asset holdings $a_{i,0}$, productivity $z_{i,0}$, ages $j_{i,0}$, durable goods $\phi_{i,0}$, securitized loans $m_{i,0}$, the law of motion for z as in equation 9 and a set of policies $\Omega_t : \{\tau_{c,t}, \mathcal{T}_t, s_{E_R,t}, s_{d_G,t}\}$, equilibrium is defined by a set of prices $P_t : \{r_t, w_t, P_t^E, P^{FF}, P_t^{d_G}, P^{d_B}\}$ and distribution μ_t over $\mathbf{x} = (a, \phi, m, y, j)$ such that

1. Households solve the recursive problem as laid out in equations 10-18, and the solution is given by value functions $V_j^{NA}(a, \phi, m, y; P_t, \Omega_t)$, $V_j^{electrify}(a, \phi, m, y; P_t, \Omega_t)$ and policy functions $a_j'(a, \phi, m, y; P_t, \Omega_t)$, $c_j(a, \phi, m, y; P_t, \Omega_t)$, $e_j(a, \phi, m, y; P_t, \Omega_t)$, $\phi_j'(a, \phi, m, y; P_t, \Omega_t)$ and $m_j'(a, \phi, m, y; P_t, \Omega_t)$.

2. The final-output good firm maximizes profits as defined in equation 21, electricity firm maximizes profits as defined in equation 25, and the durable good firms maximize profits as defined in equations 28 and 29.

3. All financial wealth bequeathed and all durables bequeathed equal the initial, aggregate wealth of the new-born generation, that is, for wealth

$$\int a'(\mathbf{x}) d\mu_{J-1,t} = \int a d\mu_{0,t}, \tag{31}$$

durable goods are passed on, such that the joint distribution of durables and income is identical in the dying and newborn generation.

4. The government budget constraint holds as defined in equation 30.

5. Prices are such that markets clear for

$$\text{Capital:} \quad \sum_{j=0}^{J-1} \int a_t d\mu_j = K_{C,t} + K_{E_R,t} + K_{E_F,t} \tag{32}$$

$$\text{Labor:} \quad \sum_{j=0}^{J-1} \int y d\mu_{j,t} = L_C \tag{33}$$

$$\text{Gr. Durable:} \quad Y_t^{d_G} = \sum_{j=0}^{J-1} \int \phi' \delta_{d_G} + (\phi' - \phi) d\mu_{j,t} \tag{34}$$

$$\text{Br. Durable:} \quad Y_t^{d_B} = \sum_{j=0}^{J-1} \int (1 - \phi') \delta_{d_B} - \chi(\phi' - \phi) d\mu_{j,t} \tag{35}$$

$$\text{Electricity:} \quad Y_t^{E} = \sum_{j=0}^{J-1} \int e \phi' d\mu_{j,t} + E_{C,t} \tag{36}$$

and the aggregate resource constraint clears by Walras's law.[4]

3 Parametrization

To assess the quantitative trade-offs behind climate policy during the green transition, I parametrize the economy so that its initial steady state matches the pre-transition period. I focus on the European green transition, and calibrate to the year 2015, before the Paris

[4]The aggregate resource constraint includes the loan interest payments made by households to some foreign investors as these loans do not affect capital markets.

agreement and when electrification rates were still very low. In the initial steady state, I assume the green durable is not available yet.

Demographics and Preferences Households are born at age 25, work until 65 and die at 75. This implies J is 50, and J_w is 40. Generations are equally sized. I set the intertemporal elasticity of substitution $1/\sigma$ to $1/2$, a standard value in the life cycle literature. I calibrate the remaining preferences parameters: discount factor, β, consumption share, γ, energy subsistence level, $\bar{\mathcal{S}}$, bequest preference, η, and bequest luxury, $\underline{b}$, internally, using a simulated method of moments. The key moments I target to discipline these parameters are the mean savings in the economy, the average energy expenditure over total expenditure, the degree of non-homotheticity in energy expenditures, the change in saving holdings during the retirement phase and inequality in bequests. I discuss this in more detail in section 3.1.

Durable good and Energy The composite durable converts energy into heat and mobility services. I follow Kuhn & Schlattmann (2024) in defining two-thirds of the composite good as the car and the remaining part as the heating system, based on expenditure shares on energy for both components. I define the rate at which energy is translated into services, $A_{s,G}$ and $A_{s,B}$ for the green and brown durable good respectively, as energy efficiency. I calibrate the efficiency levels of the two technologies separately for heat and for mobility, then aggregate them for the composite good.

Energy efficiency is the rate at which a unit of energy, fossil fuel or electricity, translates into useful energy by the technology at hand: how much useful energy a technology extracts from its primary (or, for electricity, secondary) energy source. Due to transformation losses or gains at the time of use, energy services consumed are not equal to final energy demand.

For heating, I use energy efficiency estimates from IEA (2022).[5] They define useful energy as energy available to satisfy the needs of the user, and refer to this as energy services demand, similar to my model. In their report, they call the difference between useful energy and the energy used to run the technology as the coefficient of performance (COP), which directly translates into $A^H_{s,G}$ and $A^H_{s,B}$ in my model. They state that heat pumps have a COP of around 4, while the range is wide: industrial heat pumps have a COP around 3, and district heating networks with large heat pumps reach a COP of 8. I set $A^H_{s,G}$ to 4. Natural gas boilers, on the other hand, exhibit a COP of around 0.8, so I set $A^H_{s,B}$ to 0.8.

[5] The Future of Heatpumps report

Parameter	Description	Value	Internal	Target/source
Households				
Preferences				
β	Discount factor	0.945	yes	See table 2
$1/\sigma$	Intertemporal substitution	1/2	no	Standard value
γ	Taste for consumption	0.916	yes	See table 2
$\bar{\mathcal{S}}$	Energy service subsistence	7.98	yes	See table 2
η	Strength of bequest motive	33.76	yes	See table 2
$\underline{b}$	Luxury of bequest	1.8	yes	See table 2
Durable goods and energy				
$A_{s,G}$	Energy efficiency d_G	1.82	no	See text
$A_{s,B}$	Energy efficiency d_B	0.47	no	See text
δ_G, δ_B	Depreciation	0.02, 0.01	no	Schloter (2022)
χ	Resale value	0.27	no	Gilmore & Lave (2013)
λ_{LTV}	Loan constraint	0.9	no	Own assumption
r_m	Loan premium	0.03	no	Realistic risk premium
μ_{adj}	Location adjustment shock	-1.82	yes	See table 2
σ_{adj}	Scale adjustment shock	0.38	yes	See table 2
Income and wealth				
ρ_z	Persistence	0.98	no	DNB Household Survey
σ_ϵ	Std. dev. income	0.1	no	DNB Household Survey
σ_{a_0}	Std. dev. wealth at age 0	0.556	no	DNB Household Survey
p_0	Zero-wealth share at age 0	0.39	no	DNB Household Survey
Production				
Electricity				
α_R	Renewable share	0.5	no	see text
λ	EoS between E_R and E_F	1.8	no	Papageorgiou et al. (2017)
ζ	Capital share E_F	0.3	no	Own assumption
ω	EoS between K_{E_F} and FF_E	0.25	no	Coenen et al. (2024)
A_{KR}	Productivity green capital	15.43	yes	See table 2
A_{KF}	Productivity brown capital	17.82	yes	See table 2
Durable goods				
A_B	Productivity Y^{d_B}	2.2	no	Price of d_B
A_G	Productivity Y^{d_G}	0.86	no	Price gap, see text
Final consumption good				
α_c	Capital share	0.33	no	Literature
ν	EoS capital-labor and energy	0.2	no	Realistically low, see text
ψ	Energy share	0.1	no	See text
ι	EoS between E_C and FF_C	1.5	no	Papageorgiou et al. (2017)
κ	Electricity share	0.2	no	Papageorgiou et al. (2017)
A_{KC}	Productivity capital-labor	1.034	yes	Normalization w
A_{EC}	Productivity electricity	0.018	yes	See table 2
A_{FFC}	Productivity fossil fuels	0.00059	yes	See table 2

Table 1: Parameter values

For mobility, I use estimates from the US Department of Energy from 2024.[6] They find that for electric vehicles, the energy conversion rate from energy input to useful energy, $A^M_{s,G}$, lies around 0.75.[7] For a gasoline car, energy efficiency $A^M_{s,B}$ is lower, as it only translates a fraction 0.3 of total energy into useful energy and the remainder is lost as heat. This adds up to $(0.33 \times 4 + 0.67 \times 0.75 =)1.82$ for $A_{s,G}$, and $(0.33 \times 0.8 + 0.67 \times 0.3 =)0.47$ for $A_{s,B}$.

For other parameters, I follow the empirical literature. Following estimates from Schloter (2022), electric vehicles depreciate faster than gasoline vehicles. Estimates for heat pumps are diverging. I set the depreciation for the green durable good, δ_{d_G} to 2%, against 1% for the brown durable good, δ_{d_B}. For the resale value, χ, I follow a resale value for cars from Gilmore & Lave (2013) of 40% and 0% for heating systems; for the composite good, this implies a resale value of 27%.

I calculate the relative (to income) price of fossil fuels as a composite of the gasoline price and the natural gas price, for mobility and heating purposes respectively, equal to 0.058 euros per kWh, or 0.00223 in model units.[8] The price of one unit of fossil fuels then pins down the tax in percentage to price per tCO2 unit, as there is a proportional mapping between emissions and units of fossil fuels.

For the parameters governing secured borrowing for the purchase of a durable good, I set the minimum down-payment requirement at 10%, a realistic value I assume directly. I also add a premium on loans to households, since such loans typically carry high interest rates.

Income and initial wealth Income is determined by the endogenous wage w and exogenous, idiosyncratic productivity $y_{i,j}$. Productivity splits into a deterministic life cycle and a stochastic component. I normalize average productivity such that the average productivity equals one. This implies an average net labor income of 26033 euros in 2015.

I calibrate life cycle component ξ using the DNB household survey, by regressing the log of net income on age dummies and year fixed effects. Figure 1 shows the smoothed life cycle pattern of productivity. I then use the regression residual to calibrate the idiosyncratic, stochastic part $z_{i,j}$. Appendix A gives further details on the data used to calibrate of the income process.

[6] https://www.energy.gov/eere/vehicles/articles/fotw-1360-sept-16-2024-typical-ev-87-91-efficient-compared-30-conventional

[7] taking into account cars with and without regenerative braking

[8] For natural gas, Eurostat reports a price of 0.053 euros per kWh in Europe in 2015. (Online data code: nrg_pc_202) I take the price of Euro-super95 from weekly oil bulletin. This price varies significantly year-to-year, to be on the safe side, I take the average over 2014-2016. The price of gasoline is then 532 euros per 1,000 liters. Using a conversion of 8.9 kWh per liter of gasoline, I calculate the price of gasoline at 0.06 euros per kWh in Europe in 2015. Because the composite good is 67% mobility, the price of fossil fuels is 0.058 euros per kWh. I define the model unit as 1,000 kWh and normalize it by mean income, which sets the price in the model to 0.00223.

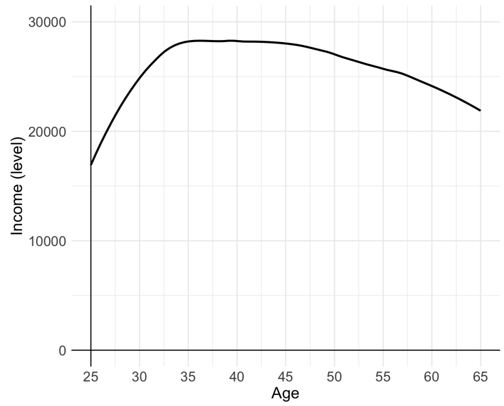


Figure 1: **Median income** Deterministic life cycle component of productivity, $\xi(j)$

For wealth data I again rely on the DNB household survey. Initial wealth is a combination of a degenerate distribution at zero wealth and a log-normal distribution over non-zero support, independent of income. I calibrate the fraction of households with a zero level of wealth, p_0, as those households with no or negative financial assets in the data at age 25. For the log-normal distribution, the mean is determined endogenously so that the value of assets held by the newborns integrates up to the value bequeathed by the dying households. I calculate the variance of the log normal distribution, $\sigma^2_{a_0}$, as the variance across all observations with positive financial assets at age 25.

Electricity firm I assume α_R, the renewable electricity share in the aggregator, to be 0.5. This parameter is inconsequential, as I calibrate the productivity internally to target the renewable electricity share (see table 2). Choosing a different value will not affect the results. For the elasticities of substitution, I use estimates from the empirical literature: 1.8 for the elasticity of substitution between green and brown sources in electricity generation λ, from Papageorgiou et al. (2017) and 0.25 for the elasticity between capital and fossil fuels in brown electricity production, ω, from Coenen et al. (2024). There is no consensus in the literature on the remaining parameter, ζ, which governs the capital share in the production of E_F, and I set it to 0.3. Again, as I internally calibrate the productivity of capital in E_F, this parameter is inconsequential.

I choose the productivity of green capital A_{KR} and brown capital A_{KF} in the electricity sector to target (1) the share of renewable electricity and (2) the relative price of electricity to fossil fuels, described in section 3.1.

Durable good firms I set the productivity level of the firm producing the brown durable good, A_B, such that the price of the brown durable good equals nearly half of the mean yearly income, namely 12000 euros. The green durable good becomes available at the start of the transition. Its productivity is set such that the price premium between the green and brown durable is 2.55. This value is informed by the relative investment cost of electric vehicles and heat pumps in 2015. Although the precise price gap is difficult to determine, some estimates from the literature can be used.[9] For cars, Lévay et al. (2017) report matched-model net purchase-price ratios spanning roughly 1.16–2.65 across the BEVs in their sample. In line with that evidence, Weiss et al. (2019) find a performance-adjusted price ratio of 1.82 for 2016. For heat pumps this price gap is larger and with a wider variance between countries, around 3 to 5 at that time, so I choose an average of 4. The convex combination of the two durables gives a price gap of 2.55. As the measurement of the price gap is noisy but an important determinant of switching behavior, I assess the robustness of results against a smaller premium in section 5.

Final consumption good firm I follow the literature for both the capital share in the capital-labor composite α_c, and the elasticity of substitution, ν, between capital-labor and energy. For the latter, the literature agrees on a realistically low elasticity of substitution ranging from 0.1 (Hochmuth et al., 2025; Fried et al., 2022) to 0.2 (Olovsson & Vestin, 2023). To stay conservative on the tax required to reduce industry emissions while remaining within the accepted range, I set ν to 0.2. Part of the results below depend on the fact that this parameter is below one. The literature agrees on that this parameter should be below one.

For the energy share, ψ, Hassler et al. (2021) show that this parameter is unimportant in this specification of the firm problem so long as ν is not close to one. For this parameter, Hochmuth et al. (2025) assume a value of 0.05, and Ascari et al. (2025) discipline it internally. Again, the parameter does not move much as the productivities are calibrated internally conditional on it. I set this value to 0.10 similar to the low estimates of energy expenditures out of total expenditures in the European economies. To pin down the elasticity of substitution between the different fuel types, ι, I use the empirical estimate from Papageorgiou et al. (2017), who estimate ι and the electricity share, κ, at 1.5 and 0.2 respectively.

I internally calibrate the rates at which energy is transformed into the final good, productivities A_{EC} and A_{FFC}, to target the electricity input share and the share of total fossil fuels used in industry in section 3.1. Lastly, I set the productivity of capital-labor to normalize the wage to one in the initial steady state.

[9] For example, looking at the average price of purchased cars in that year might already include a bias due to self-selection in type of car, and price is correlated with size, luxury or other car characteristics.

3.1 Internally calibrated parameters

To discipline 11 parameters that lack a clear data counterpart, I target 11 implied moments in the simulated data from the model solution. There is not a one-to-one relation between the parameters and moments in the calibration, however I choose 11 moments which are directly related to one of the parameters to get a good calibration. I proceed in two steps. I first calibrate 9 parameters that affect behavior in the steady state. I choose these 9 parameters, $\theta = (\gamma, \bar{\mathcal{S}}, \beta, \eta, \underline{b}, A_{KR}, A_{KF}, A_{EC}, A_{FFC})$ in order to minimize the objective function

$$\min_{\theta} \sum_{i=1}^{9} W_i \left(\frac{M_i^{model}(\theta) - M_i^{data}}{M_i^{data}} \right)^2, \tag{37}$$

in which W_i gives a weight for each moment i. Most weight is put on the parameters governing the mechanism explicitly; i.e. the utility parameters governing energy consumption $(\gamma, \bar{\mathcal{S}})$ and electricity sector productivities (A_{KR}, A_{KF}). After having calibrated the initial steady state, there are two parameters left open, μ_{adj} and σ_{adj}. These govern the i.i.d. adjustment shifter, $\mathcal{E}_{adj}$, that determines the probability of electrification as a function of the value function differential. Because the green durable good is not available in the initial steady state, these parameters do not affect the initial steady state. Conditional on the initial steady state calibration, I discipline the 2 parameters by solving for the entire transition path, and targeting moments from the early transition. Table 2 shows the 11 moments targeted, the parameter most directly related to each moment, how the moment is calculated in the model simulation, the data target, M_i^{data}, and the result from the model simulation, M_i^{model}.

In order to discipline γ, the taste for consumption in the utility function, and $\bar{\mathcal{S}}$, the subsistence level for energy services, I target moments that are related to the energy expenditure share of households. I follow Hochmuth et al. (2025), and use the Household Budget Survey on income, consumption and wealth from Eurostat for 2015. Using household-level data, they calculate expenditure shares for a variety of goods as well as total expenditures for each income quintile. I use the data to calculate the energy expenditure share over total expenditure (defined as expenditures on all HICP items) for each income quintile. In figure 2 the dotted gray line shows the energy expenditure share for households in each income quintile in the data. In the calibration, I target the average energy expenditure share of 12.5% and the degree of non-homotheticity measured by the energy expenditure share in the first over the last quintile, equal to 1.6. Solving the model allows me to calculate the energy expenditure share for each income quintile in the model, which is shown by the red line in figure 2.

I discipline parameters β, η, $\underline{b}$, respectively the discount factor, bequest strength, and

	Moment	Calculation	Data	Model
γ	Mean energy expenditure share	$\int_{i,j} P^{\phi_{i,j}} e_{i,j}/(P^{\phi_{i,j}} e_{i,j} + c_{i,j}) d\mu$	0.125	0.125
$\bar{\mathcal{S}}$	Non-homotheticity	$\int_{i,j\|Q1} \frac{P^{\phi_{i,j}} e_{i,j}}{P^{\phi_{i,j}} e_{i,j} + c_{i,j}} d\mu \Big/ \int_{i,j\|Q5} \frac{P^{\phi_{i,j}} e_{i,j}}{P^{\phi_{i,j}} e_{i,j} + c_{i,j}} d\mu$	1.6	1.6
β	Mean wealth over mean income	$\int_{i,j} a_{i,j} d\mu \Big/ \int_{i,j} w y_{i,j} d\mu$	3.6	4.33
η	Decay of wealth	$\int_{i} a_{i,J} d\mu \Big/ \int_{i} a_{i,J_W} d\mu$	0.83	0.69
$\underline{b}$	Wealth inequality at death	$\int_{i\|D6} a_{i,J} d\mu \Big/ \int_{i\|D3} a_{i,J} d\mu$	2.84	2.28
A_{KR}	Green electricity share	$\frac{E_R}{E_R + E_F}$	0.175	0.175
A_{KF}	Relative price of electricity	P_E/P_{FF}	2.37	2.33
A_{EC}	Input share electricity final good	$E_C/(E_C + FF_C)$	0.34	0.28
A_{FFC}	Final good fossil fuel share	$FF_C/(FF_C + FF_{HH} + FF_E)$	0.46	0.44
μ_{Adj}	Percentage electrified 2017	$100 * \int \phi \, d\mu_{2017}$	1.4	1.33
σ_{Adj}	Price elasticity d_G 2017	$\frac{\partial \int \phi \, d\mu_{2017}}{\partial P^{d_G}} \frac{P^{d_G}}{\int \phi \, d\mu_{2017}}$	-1.36	-1.35

Table 2: Target moments in internal calibration

bequest luxury by targeting moments related to wealth and income. These moments are calculated from the data using the DNB household survey, waves 2010 - 2020. I discuss this dataset in appendix A. For the discount factor β, I target the ratio of mean household financial net worth (NW) to mean net household income. Second, for the strength of the bequest motive, η, I measure the mean decay of net worth in the data as the mean net worth at age 75 over the mean worth at age 65, in order to target the ages of retirement and death in the model. Lastly, to discipline the degree of luxury of bequests, $\underline{b}$ I target the degree of bequest inequality. I compute the mean financial net worth in the third and sixth decile at age 75, and take the ratio as the measure of inequality.

To discipline the productivities in the model, I use aggregate Eurostat data. I set the productivity of the green capital in electricity generation, A_{KR} to target the share of electricity that is produced from renewable sources, which in Europe in 2015 is 17.5%.[10] To determine the productivity level of the brown capital in electricity generation, A_{KF}, I target the relative price of electricity to fossil fuels. As explained above, the price of fossil fuels is 0.058 per kWh. For the electricity price, I use the Eurostat DC-band price per kWh in

[10]Online data code: nrg_ind_ren

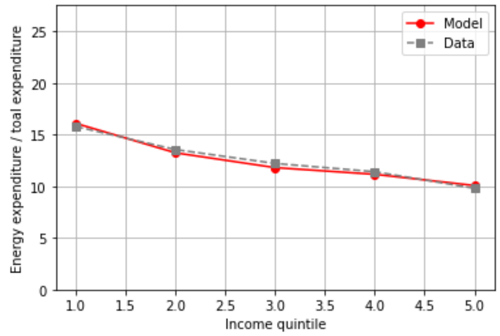


Figure 2: **Energy expenditures** Energy expenditure share per income quintile in Eurostat data (2015) and in the model initial steady state.

Europe in 2015, which is 0.1372 euros.[11]

I discipline the productivities for both energy sources in the production of the final consumption good, A_{EC} and A_{FFC} by targeting the input share of electricity in the energy mix and the share of total fossil fuels used in industry. Eurostat aggregate data for industry excludes the electricity sector but include all energy used by sectors that carry out other manufacturing and industrial processes.[12] Electricity makes up 34% of all energy used in 2015 in the EU. To determine the share of total fossil fuels that is used in production, I follow calculations from the Eurostat energy statistics report.[13] Of all fossil fuels used in the European economy, 46% went to industry (defined as manufacturing, storage, and services). Further fossil fuel consumption happens in electricity production (21.8%) and by households for heating and mobility (26.2%).[14]

Lastly, in the second calibration step, the two remaining parameters, μ_{adj} and σ_{adj} are disciplined by targeting moments that result in the first years of the transition; they govern the aggregate rate of, and the heterogeneity within, households switching from a brown to a green durable. I discipline the location and scale of the extreme value shock, μ_{adj} and σ_{adj}, using the strategy from Kuhn & Schlattmann (2024). I target the share of households that switched to a green durable, and the own-price elasticity of the green durable good in the third year of the transition, 2017. From Statistics Netherlands (CBS), the share of households with an electric vehicle in 2017 is 1.4%. The share of households with a heat pump in 2018 is 1.54%. For 2017, I target 1.4%. The model slightly underestimates the aggregate share of adopters in 2017. However, as I show in section 3.2, it does match the overall time trend well, and the underestimation is only in the highest income group. This

[11]Online data code: nrg_pc_204, euro area
[12]Online data code nrg_bal_s
[13]Energy accounts, 2025
[14]The remaining 6% is used for, among other things, mining, water supply, and construction.

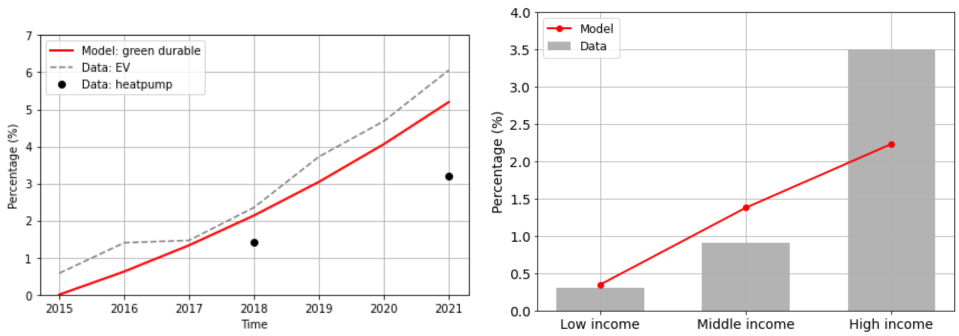


Figure 3: **Electrification behavior** Left: Share of households that has electrified. Right: cross-section of households who have electrified in 2017.

alleviates concerns about claims for the low-income households. For the elasticity, I rely on estimates of Fridstrøm & Østli (2021) who use Norwegian data to estimate the price elasticity of cars. In 2016, they find elasticities of -0.99 and -1.72 for battery and plug-in hybrid electric cars. As in Kuhn & Schlattmann (2024), the average elasticity is -1.36.

3.2 Baseline model validation

To validate the model, I evaluate the baseline switching behavior in the first years of the transition, and I compare inequality in the model and data. I focus on these two characteristics because they represent or relate to the main mechanism (electrification) and the main outcome variable (inequality).

Switching behavior I use publicly available Dutch data from CBS on the share of homeowners with a heat pump and car owners with an electric vehicle. As in the model, the durable good is a composite of both, so I examine the realized time trends for each in the aggregate. Figure 3, panel (a), shows an overall good fit for uptake in the first years of the transition. 2015 is the year of introduction, so uptake starts at zero by assumption. The internal calibration targets the aggregate share of the composite good in 2017; the other years are untargeted.

Additionally, I analyze the heterogeneity in electrification across three income groups in 2017. I take a linear combination of the share of households with a heatpump (33%) and the share of households with an electric vehicle (67%) conditional on income: the lowest and highest quartiles, and the middle half. The percentages in the linear combination follow from the definition in the durable good. For heat pump shares, I use micro data from CBS for 2018, explained and shown in appendix A.1. For the uptake of electric vehicles, I use estimates from CBS, who use administrative data to calculate the share of EV owners in each income quartile.[15] They find that in 2021, out of all EV owners, 5.5% is in the lowest income quartile, and 62.4% is in the highest income quartile. I use this data from 2021 to calculate shares per income group, and multiply them with the aggregate in 2017. To get the share of EV-owners per income group I use Bayes inversion.[16] The results are shown in figure 3, panel (b). Overall, the goodness of fit of electrification for different income groups depends on the income group. The model underestimates uptake in the highest income group, and overestimates it in the middle-income group. The model's electrification pattern is more linear in income than in the data, where the relationship appears more convex. However, the uptake among low income households fits the data well which is most important for the results that follow.

Inequality As I compare the distributional welfare effects among different income groups, it is necessary to assess income inequality. In the euro area the disposable income gini ranges from 23.7 to 37.9.[17] I obtain a Gini coefficient of 36.8, at the upper end of this range.

4 Results

I solve the model using the algorithms in online appendices A - C. The economy starts in the initial steady state and unexpectedly enters a transition. In the baseline (no tax) transition, the green durable good becomes available at a price premium and the productivities of production of the green durable good A_G and green electricity A_{KR} increase.[18] I compare this baseline case to four policy experiments in which I introduce a carbon tax at the start of the transition, phased in linearly over 25 years. At the start of the transition, households learn the entire path of productivities, carbon tax, recycling instruments and prices.

In the policy experiments, I increase the carbon tax $\tau_{c,t}$ from zero in time $t = 0$ to $\bar{\tau}_c$ in 25 years, and keep it fixed afterwards. I consider four revenue-recycling instruments: (i) government expenditure, G, (ii) a uniform lump-sum transfer, $\mathcal{T}$, (iii) a subsidy for green electricity production, s_{E_R}, and (iv) a subsidy for electrification of households, s_{d_G}, which I cap at 30%.[19] For each instrument, I compute the tax rate $\bar{\tau}_c$ that achieves a maximum

[15] CBS report: who owns electric vehicles

[16] Q1: $0.055 \times 0.014/0.25 = 0.31\%$, Q2-Q3: $0.321 \times 0.014/0.50 = 0.90\%$, Q4: $0.624 \times 0.014/0.25 = 3.49\%$.

[17] Eurostat, 2015

[18] I extrapolate the trends in price developments over the past 10 years and assign the same growth factor for the first 15 years of the transition. For the green durable productivity I make sure it converges to the brown durable's productivity A_B.

[19] I do this as the revenue including the firms' tax payments exceeds the amount that can be given out as subsidies (that is tax revenues exceed all switching costs in early periods even if everyone were to switch). To avoid taxing 100% of the switching cost, I set the maximum rate to 30%, in line with common subsidies given out in Europe. In the default case, the remainder is spent on government expenditure. Distributing

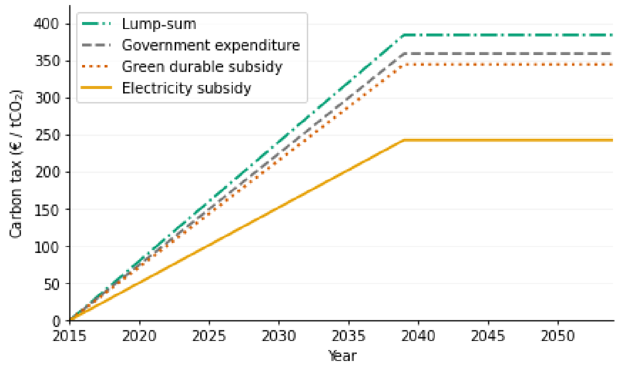


Figure 4: **Carbon tax** Path of carbon taxes per revenue-recycling instrument to meet the exogenous temperature target.

temperature increase of 2 degrees Celsius above the pre-industrial level in 2065.[20] I compute the development of temperature as a function of total cumulative emissions, using an off-the-shelf model discussed in appendix B.1 from Douenne et al. (2025a), mapped to model units in appendix B.2, exactly taken from Douenne et al. (2025b). I show the development of temperature in appendix B.3. By calculating an endogenous tax rate which reaches a certain temperature, climate damages are equal between the different instruments. Therefore, I can compare outcomes between instruments without modeling damages explicitly.[21]

Before examining the efficiency and distributional effects of each policy, I first highlight in figure 4 the endogenous tax paths for each instrument needed to achieve the temperature target. Using a green electricity subsidy reduces the peak tax rate required to achieve the temperature target. The reason for the lower tax rate with a green electricity subsidy lies in two mechanisms. First, with a subsidy for green electricity, for a given tax rate, the electricity sector substitutes more towards renewable generated electricity, so electricity production is less emission-intensive. Second, the subsidy for renewable electricity production lowers the electricity price compared to other instruments and even the baseline, so for a given carbon tax, the final consumption good producer substitutes more toward electricity, away from fossil fuels within the energy composite, M. The green durable subsidy has a smaller tax-reducing effect because households at the margin of switching in the baseline case are mostly poor and use little energy overall. Lump-sum increases the tax rate (compared to G) as it increases consumption of the poor.

the remainder on electricity subsidies barely affects results.

[20] As the economy is not in net-zero at this point, the underlying assumption is that some carbon capture technology keeps the temperature at 2 degrees Celsius.

[21] Alternatively, I could set an exogenous tax rate and compute climate damages. In this case, climate damages would be lower with a subsidy rather than lump-sum redistribution or government expenditure due to lower emissions, which reduces GDP declines. In the current case I account for this by allowing for a lower tax rate under subsidy redistribution, also leading to lower GDP declines.

(a) GDP (b) Wage

(c) Interest rate (d) Electricity price

Figure 5: **Efficiency** GDP and prices over the transition.

4.1 Efficiency effects

To determine the extent to which the carbon tax distorts the economy, I assess the development of GDP and prices over the transition. Figure 5 first shows the baseline transition in black. In the absence of climate policy, GDP and the wage gradually increase, and the electricity price decreases over the transition due to the rising productivities. The interest rate initially increases and subsequently declines. Early in the transition, capital demand rises because production shifts toward the capital-intensive electricity sector when its productivity rises. At the same time, households use their savings to purchase green durables rather than holding it. Capital demand therefore increases relative to supply, pushing up the interest rate. This is temporary: once households have electrified, they increase their financial wealth, causing the equilibrium interest rate to decline. Figure 5, panel (a), shows that GDP initially declines after the climate policy is introduced, for all instruments, because higher energy prices raise firms' cost of production.[22] Relative to the baseline, GDP remains lower throughout the transition.[23] The recovery of GDP after the initial decline

[22]Discontinuities in the graph are explained by the phasing in of increased renewable electricity production for 15 periods (until 2030), and the tax for 25 periods (until 2040).

[23]Importantly, these GDP losses capture the economic costs of reducing emissions, as I abstract from the economic damages caused by climate change. They should therefore not be interpreted as the net effect of

depends heavily on the instrument used. Under lump-sum transfers, green-durable subsidies, and government spending, GDP follows a similar path and remains substantially below its baseline level. Under the electricity subsidy, by contrast, GDP begins to recover more strongly after 2040, which is caused by lower energy price increases. As shown in figure 4, the required tax rate is significantly lower for the electricity subsidy, causing fossil fuel prices to rise less. Furthermore, figure 5, panel (d) shows that the electricity price significantly decreases, even more than in the baseline, when the production of renewable electricity is subsidized.

Relative to the baseline, climate policy also increases the interest rate (figure 5, panel (c)) and decreases the wage (figure 5, panel (b)). A carbon tax shifts production from fossil energy toward electricity, which is more capital-intensive, raising capital demand and pushing up the interest rate. The wage drops because the capital-labor bundle is complementary to the energy composite $\nu < 1$: as energy prices rise, firms demand less of both. Under the electricity subsidy, the wage drops less and the interest rate spikes more. The muted wage drop follows from the smaller increase in the price of M — fossil fuels cost less because the required tax is lower, and the electricity price falls rather than rises. The larger interest-rate spike reflects stronger substitution toward electricity in final-good production.

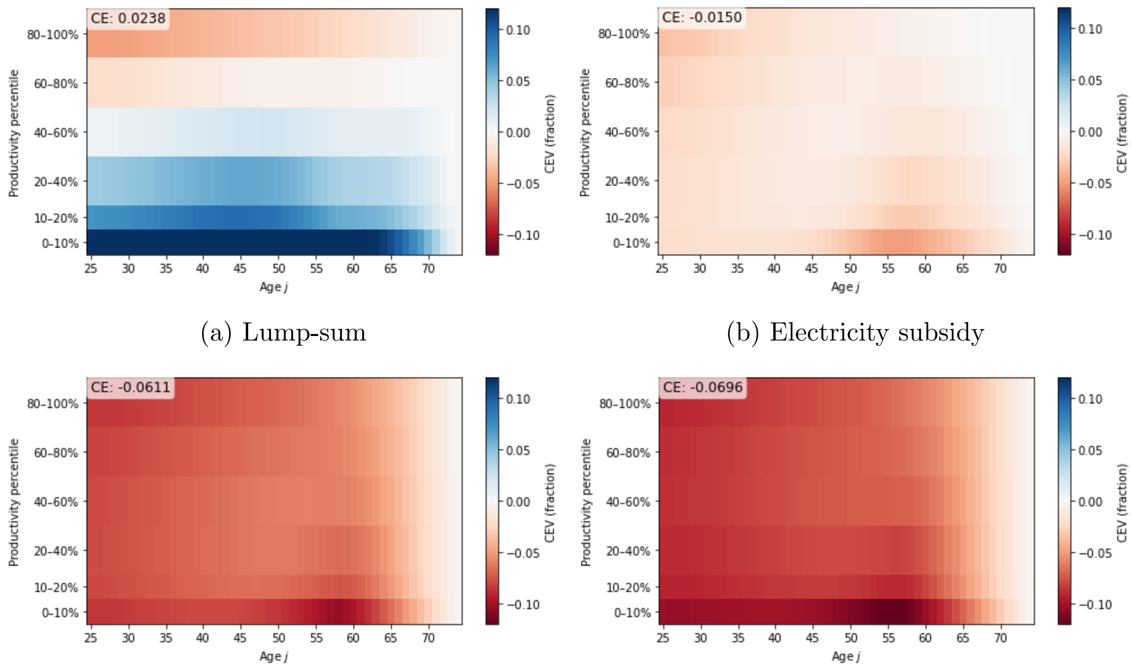
4.2 Equity effects and the electrification margin

In figure 6 I show the welfare effect of the carbon tax, calculated as the consumption equivalence, for the average household in a given age and productivity group (j, z), for all households alive when the policy is introduced.[24] Aggregate welfare is the average consumption equivalent. The figure shows that, with the exception of uniform lump-sum transfers as the recycling instrument, all agents are worse off from the carbon tax, absent any climate damages. Under lump-sum transfers, low-income households gain because the transfer raises their net disposable income, even after accounting for lower labor income, by more than their expenditures rise.

In all three other cases all agents are worse off with carbon tax, and poor, middle-aged households suffer the largest welfare loss. Subsidies increase welfare compared to using tax revenues for wasteful government expenditure. When the subsidy targets renewable electricity, the welfare losses are smaller overall than under a green durable subsidy. This welfare difference can be explained by weaker negative general equilibrium effects under the electricity subsidy: as explained above, the carbon tax and the price of electricity are

climate policy on GDP once avoided climate damages are taken into account.

[24]This is the amount of consumption that the household needs ($CE > 0$) or is willing to give up ($CE < 0$) in the no-policy baseline transition in order to be indifferent between the policy and no-policy transition.



(a) Lump-sum

(b) Electricity subsidy

(c) Green durable subsidy

(d) Government spending

Figure 6: **Consumption equivalence (CE)** Welfare effect from the carbon tax across the age–income space for different recycling instruments. Values indicate the percentage change in consumption in the baseline required for households to be indifferent between the baseline and policy scenario. Top left number shows aggregate CE.

significantly lower following the electricity subsidy, so wages decrease significantly less for the same climate target. Together, these two effects make electricity subsidies superior to a green durable subsidy in terms of welfare.[25] However, the lower wage decrease and the drop in electricity prices mostly benefit rich, high-productivity households who electrify early. Put differently, the welfare gains of a green electricity subsidy concentrate mostly among rich households.

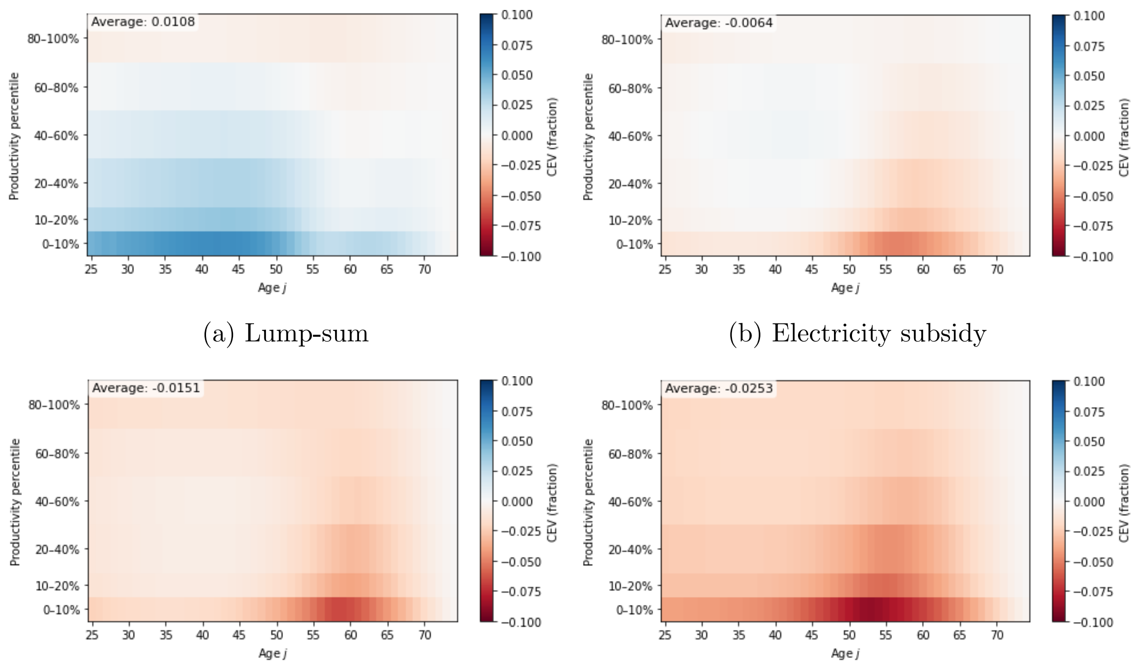
The role of electrification The electrification margin is the key driver of the high welfare loss for the low-income households. To quantify the role of the electrification margin in shaping the distributional welfare effects of the carbon tax, I compare the baseline consumption equivalent from above to a consumption equivalent from a counterfactual economy in which all households are endowed with a green durable. Removing the extensive electrification margin in this way allows me to isolate the welfare consequences it causes after a policy

[25] The combination of a green durable subsidy and electricity subsidy does not improve welfare much beyond what the electricity subsidy compared to the green durable subsidy does.

introduction. I define the state-dependent welfare effect of the electrification margin as

$$\Delta CE^{Elec}(j,z) = CE_{t=1}(j,z) - CE^{\text{noElec}}_{t=1}(j,z),$$

where in the *noElec* case all households are endowed with green durables.[26] In doing so, for each income, age group I subtract the welfare effect that would have arisen without electrification margin from the welfare effects plotted in figure 6. I plot this state-dependent welfare effect of the electrification margin in figure 7. A positive value implies that the electrification margin improves the welfare impact of the policy, while a negative value indicates that it causes a welfare loss following the policy introduction.



(a) Lump-sum
(b) Electricity subsidy
(c) Green durable subsidy
(d) Government spending

Figure 7: **Electrification margin effect** Differential in consumption equivalence with and without electrification margin, $\Delta CE^{Elec}(j,z)$, for different recycling instruments. Values indicate welfare loss or gain from the carbon tax, caused by the electrification margin.

In all cases except lump-sum redistribution, the electrification margin makes households significantly worse off from a carbon tax. In case of the electricity subsidy, the electrification margin explains 43% of the aggregate welfare loss.[27] More strikingly, for middle-aged, low-income households almost the full welfare loss is explained by the electrification margin.

[26] Taxes are not re-estimated but prices are in order to make sure that markets clear. The former is not a large concern for emissions as households' direct fossil fuel consumption of non-switchers is low.

[27] For government expenditure this is lower, 36%, because the wage drop accounts for a larger fraction. With electrification subsidies it is even lower than that, 25%, because the upfront investment cost is lower.

This shows that the electrification margin creates a heterogeneous impact of the carbon tax over the age distribution for lower-income groups. The hump-shaped pattern in the carbon tax incidence over the age distribution can be explained by an increase in expenditures on either energy, or durables. To show this, I explain the mechanisms in terms of a change in expected lifetime consumption following the policy introduction. Note that under lump-sum redistribution, the electrification friction actually makes poor households slightly better off. This happens because, with the electrification margin, tax revenues and thus transfers are higher at the start of the transition.

4.2.1 Mechanisms

To understand the mechanisms underlying the distributional welfare results and the effect of the electrification margin, I compute and decompose the change in the present value of the expected lifetime consumption following the introduction of the policy, denoted by ΔLC_i. This can be interpreted as the budgetary impact of the policy on household i at the time of the policy introduction. Quantitatively, the consumption explains the sign of the welfare effect.[28] It is especially useful to compare welfare effects between policy instruments, within income groups. I compute ΔLC_i by iterating the present value budget constraint forward, which allows me to decompose changes in ΔLC_i into four components, formally defined in proposition 1.

Proposition 1. *Consider household i, aged $j_{0,i}$ at the time of the policy introduction. Define the remaining lifetime horizon as $S_i \equiv J - 1 - j_{0,i}$. Let $Q_0 \equiv 1$ and $Q_s \equiv \prod_{u=1}^{s}(1 + r_u)^{-1}$ denote the discount factor, so that all quantities below are expressed in units of consumption at $s = 0$. For $s = 0, 1, \dots, S_i$, define household i's lifetime consumption object as*

$$LC_i \equiv E_0 \sum_{s=0}^{S_i} Q_s\, c_{i,j_{0,i}+s,s} + Q_{S_i}\, a_{i,J},$$

and define the four components

$$\text{Transfers}_i \equiv \sum_{s=0}^{S_i} Q_s\, \mathcal{T}_s, \quad \text{LaborInc}_i \equiv E_0 \sum_{s=0}^{S_i} Q_s\, w_s\, y_{i,j_{0,i}+s},$$
$$\text{EnergyExp}_i \equiv E_0 \sum_{s=0}^{S_i} Q_s\, EE_{i,j_{0,i}+s,s}, \quad \text{DurableExp}_i \equiv E_0 \sum_{s=0}^{S_i} Q_s\, DE_{i,j_{0,i}+s,s},$$

[28] The welfare impact of this consumption change depends on sign and size of the change, the timing of the change (as the future is discounted), the degree to which the change is spread out over time, and the initial level of consumption. The concavity of the value function causes the latter two effects. For these reasons, the magnitude of the change in consumption translates nonlinearly into the magnitude of welfare effects.

where

$$EE_{i,j,t} \equiv P_t^{\phi_{i,j,t}}\, e_{i,j,t},$$
$$DE_{i,j,t} \equiv P_t^{d_{\phi_{i,j,t}}} \delta_{\phi_{i,j,t},t} + \pi(j,t)\, m_{i,j,t} - \left(\phi'_{i,j,t} - \phi_{i,j,t}\right)\left(m'_{i,j,t} - (1 - s_{d_G,t}) P_t^{d_G} + \chi P^{d_B}\right).$$

Then the change in household i's expected lifetime consumption relative to the baseline admits the decomposition

$$\Delta LC_i = \Delta \text{Transfers}_i + \Delta \text{LaborInc}_i - \Delta \text{EnergyExp}_i - \Delta \text{DurableExp}_i. \tag{38}$$

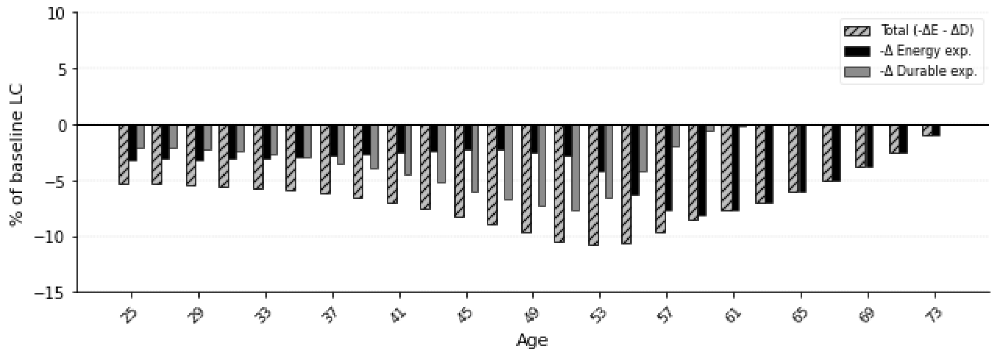
Proof. See appendix C. □

I use this decomposition to answer two questions. First, how does the electrification margin shape the distributional incidence of climate policy? Second, why does lump-sum redistribution benefit low-income households?

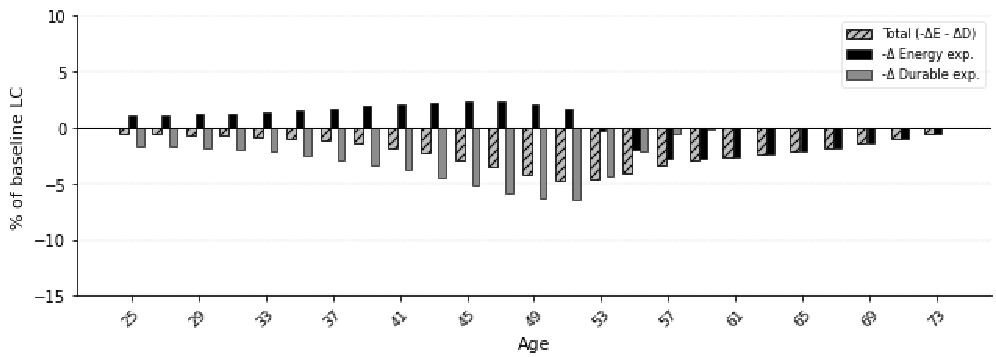
The electrification margin and age-specific expenditures To understand why the electrification margin affects the middle-aged, poor households the most, I analyze the age heterogeneity in durable and energy expenditure, as defined in proposition 1, as this heterogeneity explains the concentration of losses among the middle-aged poor figure 7. In figure 8, I plot the changes in durable expenditures, energy expenditures and their sum for the lowest income group by paired age cohort, for lump-sum and electricity-subsidy recycling.

It is useful to separate three groups of households based on their age when the transition starts, i.e. their remaining lifetime horizon: (1) the young (roughly ages 25–40) households, who would have electrified in any case; (2) the old (roughly aged > 53) households, who are not electrifying in any case; and (3) the switchers, the middle-aged (roughly ages 40–53) households, who would not have electrified without the carbon tax in place, but will do so with it. In group 1, the young households incur a small increase in durable expenditures as they electrify earlier with the policy in place, when $P_t^{d_G}$ is still higher. Under lump-sum redistribution the electricity price rises too, though proportionally less than fossil fuels, so their energy expenditures rise as well. As a result, the effect on lifetime consumption through increased expenditures is limited for this group. In group 2, because these households are too close to the end of life to recover the fixed electrification costs, they optimally stick with the brown durable and consequently face a persistent increase in energy expenditures as fossil fuel prices rise. In group 3, the policy-induced switchers, electrification requires substantial upfront costs, generating a pronounced increase in durable expenditures.

Whether these households are compensated through lower energy expenditures depends



(a) Lump-sum



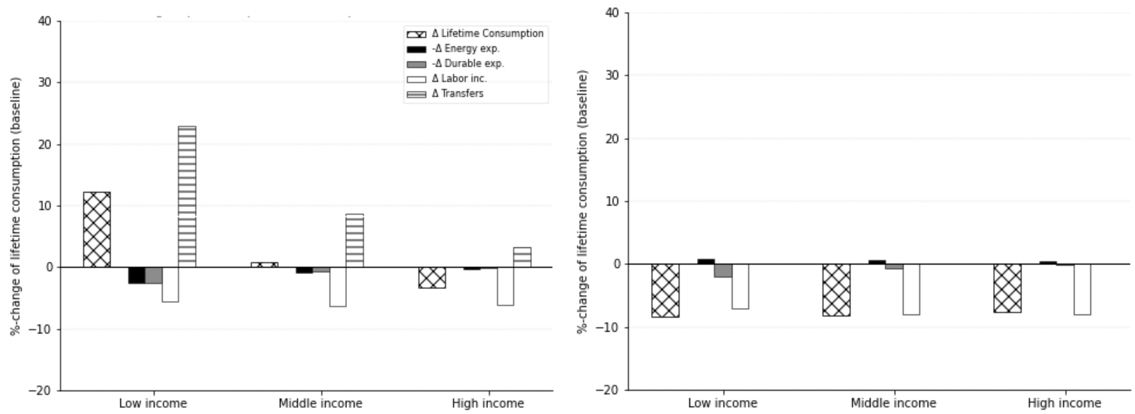
(b) Electricity subsidy

Figure 8: **Energy and durable expenditure changes** Heterogeneous expenditure responses across the age space for the lowest income group, $z = 0$.

on the recycling instrument. Under lump-sum redistribution, electricity prices rise, so switching households face both higher durable expenditures and slightly higher energy expenditures. Under an electricity subsidy, electricity prices fall, which partly offsets the switching costs through lower future energy expenditures.[29] Overall, changes in energy and durable expenditures decrease the lifetime consumption of middle-aged households the most (with the peak around ages 49–57, depending on the instrument), the ones close to the point of indifference between electrifying or not, once the carbon tax is in place. For them electrifying is as costly as not-electrifying, and as there is no clear good alternative, they are affected the most by the carbon tax. From figure 8, we learn that the electricity subsidy dampens the pain caused by the electrification margin. However, we also learned that in the case of the subsidy, middle-aged, low-income households are affected more negatively by the tax than with transfers. I now ask the question of why transfers are so beneficial for the poor.

[29] Similar figures for the green durable subsidy and government expenditure can be found in online appendix D.1. The figure for government spending explains the triangle with the same mechanisms but increased magnitudes. The figure for green durable subsidy diminishes durable spending, which explains why the triangle in figure 7 panel (c) shifts to the right and thus the margin is especially salient for never-switchers.

The benefits of lump-sum transfers To understand why lump-sum recycling benefits low-income households despite requiring a higher carbon tax than the electricity subsidy, Figure 9 decomposes the change in the discounted expected lifetime consumption, ΔLC_i, into changes in labor income, energy expenditures, durable expenditures, and transfers. Each component is expressed as a percentage of expected discounted lifetime consumption in the no-policy baseline. The denominator is therefore the same across policy instruments.[30]



(a) Lump-sum

(b) Electricity Subsidy

Figure 9: **Consumption change decomposition** Decomposition of the change in expected lifetime consumption following policy introduction. Low-income refers to the lowest decile (conditional on age), high-income refers to the highest decile.

Figure 9 shows that the largest negative component of lifetime consumption is the change in the present value of labor income. This bar reflects not a pure wage-income effect, as each object is discounted using the market discount factors of that particular equilibrium[31]. Although wages decrease less under the electricity subsidy than under lump-sum recycling, the higher interest rate discounts future labor income more strongly. This discounting effect is sufficiently large that the total present-value loss in labor income is similar for both instruments, and even slightly larger under the electricity subsidy.

The main differences between the two policies therefore arise from the expenditure and transfer components. As shown in Figure 8, the electricity subsidy performs well in limiting

[30]Similar figures for government expenditure and the electrification subsidy are reported in Online Appendix D.2.

[31]The change in discounted labor income can be decomposed exactly as

$$\Delta L_i = E_0 \sum_s Q_s^{\text{pol}} \left(w_s^{\text{pol}} - w_s^{\text{base}} \right) y_{i,j+s} + E_0 \sum_s \left(Q_s^{\text{pol}} - Q_s^{\text{base}} \right) w_s^{\text{base}} y_{i,j+s},$$

where the first term is the wage component and the second the discount-rate component.

or overturning the increase in energy and durable expenditures. Under lump-sum recycling, these expenditure burdens are larger, since both fossil-fuel and electricity prices rise. Under the electricity subsidy, switchers benefit from lower electricity prices, while non-switchers face a smaller increase in fossil-fuel prices. Nevertheless, low-income households are better off under lump-sum recycling. The reason is the transfer. The lump-sum transfer is large relative to their lifetime consumption and more than offsets the combined losses from labor income and higher energy and durable expenditures. For high-income households, the same transfer is small relative to lifetime resources and does not compensate for these losses. Their expected lifetime consumption therefore falls.

To summarize, the carbon tax raises the combined durable and energy expenditure for middle-aged, low-income households, but through different channels: switchers incur higher durable costs, while non-switchers face higher fossil-energy costs. Under renewable electricity subsidies, lower electricity prices partially offset the cost for switchers. Under lump-sum recycling, these expenditure burdens are larger, since both fossil fuel and electricity prices rise. Nevertheless, the financial relief for poor households is larger under lump-sum redistribution, given the size of the transfer relative to their lifetime consumption.

The case without electrification In contrast to the findings of Ascari et al. (2025), who show that a carbon tax combined with an electricity subsidy is regressive, I do not find regressivity with an electricity subsidy when I exclude the electrification margin for households. In their framework, regressivity arises from non-homothetic energy demand and the fact that the price of energy increases even when electricity is subsidized. The key difference is that, unlike in their setting, the price of electricity in my economy decreases when the policy is implemented with an electricity subsidy. As a result, even with non-homothetic energy demand, climate policy with revenue recycling through electricity subsidies is progressive. The reason for the declining electricity price is the assumption of continuous fuel switching of the final consumption good firm, which thus also uses fossil fuel. Because this firm pays the carbon tax on its fossil fuel usage, tax revenues increase. These additional revenues finance the renewable electricity subsidy, and the resulting subsidy-induced reduction in electricity prices more than offsets the upward pressure from the carbon tax on fossil fuels. Consequently, electricity prices fall, and policy effects turn progressive.

4.2.2 Combining lump-sum and electricity subsidies

From the above, we can establish that while a green electricity subsidy mitigates losses in GDP by lowering energy prices, it is regressive via the electrification margin. Lump-sum transfers make the carbon tax progressive at the cost of a higher GDP loss. In this section, I analyze whether combining electricity subsidies with lump-sum transfers can mitigate the trade-off between efficiency and equity. To assess this, I consider linear combinations of the two instruments, varying the share of tax revenues allocated to each. Table 3 reveals the central trade-off in this paper: subsidizing green electricity relieves GDP distortions caused by higher energy prices, but as the resulting benefits mainly accrue to richer households, aggregate welfare losses and inequality increase. Increasing the weight on lump-sum transfers raises GDP losses while improving welfare (especially for low-income households), whereas shifting revenues toward electricity subsidies reduces GDP losses but worsens distributional outcomes. Replicates of figures 5 and 6 for the combined instruments can be found in online appendix D.3.

% Tax Rev. Subsidy (LS)	Future GDP Decrease (%)	Current CE Aggregate (%)	Current CE low-inc. (%)
100 (0)	-3.41	-1.50	-2.69
75 (25)	-3.78	-0.73	2.18
50 (50)	-4.26	0.13	3.69
25 (75)	-4.90	1.12	7.90
0 (100)	-5.86	2.38	13.45

Table 3: Policy trade-off between green electricity subsidy and lump-sum transfers

Electricity subsidies primarily operate through general-equilibrium effects: lower energy prices reduce production distortions. Lump-sum transfers instead operate through direct redistribution. Because interaction effects are quantitatively small, combining the instruments does not circumvent the efficiency–equity trade-off, it mainly selects a point along an approximately linear frontier. Greater reliance on subsidies preserves output at the cost of more regressive incidence, while greater reliance on transfers improves distributional outcomes at the cost of larger aggregate distortions. Hence, the policy mix cannot deliver both the efficiency gains of electricity subsidies and the distributional benefits of lump-sum transfers. The optimal combination remains a normative question, that requires specification of a welfare function, and weights for future generations.

5 Robustness and validity

A concern about the validity of the results is that I do not model economic damages from temperature increases, which complicates the comparison between the baseline and the policy scenarios. This does not affect the comparison across policy instruments themselves, since I estimate the carbon tax path so that temperature changes do not diverge across them. Therefore, levels against the baseline omit avoided damages and are therefore lower bounds on the welfare gain from policy, while differences across instruments are damage-free by construction.

Moreover, five assumptions matter for the results and are natural candidates for robustness checks. First, idiosyncratic risk burdens households, but lump-sum redistribution partly offsets it, since deterministic transfers smooth consumption. Second, the elasticity of substitution between renewable- and fossil-generated electricity, λ, largely governs the tax rate needed to limit the temperature increase under each instrument. Third, the elasticity of substitution between energy and the capital-labor composite in the final consumption good sector, ν, sets the size of the wage decrease the tax causes. Fourth, the minimum down payment required to purchase the green durable, λ^{LTV}, governs the severity of the financing friction. Fifth, and related to the fourth, the price premium at which the green durable good becomes available, A_{d_G} directly affects household switching behavior. I assess the robustness of these assumptions and calibration choices individually.

Idiosyncratic income risk A natural source of concern arises from the fact that perfect foresight with respect to lump-sum transfers alleviates distortions from uninsurable idiosyncratic risk, and is therefore welfare enhancing. In order to test whether the result found above holds true under stricter assumptions, I solve the model for an economy with the same income states but without any risk, i.e. the transition matrix is now an identity matrix, holding everything else constant and re-analyze the welfare results. I find that both the aggregate and the distributional incidence of welfare losses remain qualitatively and quantitatively within the same magnitude under the assumption of no idiosyncratic risk. The aggregate gain in welfare from lump-sum distribution reduces from 2.38% to 2.01%, and the burden is still concentrated among the middle-aged, low-income households. This does not speak to the concern that lump-sum might increase aggregate welfare as it alleviates current inequalities. However, the conclusion I can draw, and the benefit to lump-sum that is clear compared to all other instruments is that lump-sum is the only instrument which is not regressive.

EoS renewable and fossil electricity, λ The elasticity of substitution governs the extent to which the electricity firm can substitute towards renewable sources when the carbon tax rises. A higher λ is therefore expected to lead to a lower peak tax rate, but the degree to which it does could depend on the instrument, i.e. whether or not renewable electricity is subsidized. In order to obtain an idea of the significance of this parameter, I compare tax rates in the baseline regime ($\lambda = 1.8$), with a high ($\lambda = 5$) and low ($\lambda = 0.5$) elasticity regime, in line with ranges used in Hochmuth et al. (2025). I show the estimated tax rates in online appendix E, figure 7. Results are in line with findings from Hochmuth et al. (2025), who find that this elasticity of substitution is an important determinant of the endogenous tax rate required to transition to a low-emission economy. The required tax rate varies from $670e/tCO2$ in the low-elasticity case, to $238e/tCO2$ in the high-elasticity case for lump-sum transfers. Similarly, for the renewable electricity subsidies it varies from $485e/tCO2$ to $148e/tCO2$. Importantly, as expected I do not find any interaction effects with the instruments. That is, the effect of λ on the maximum tax rate is the same for each instrument. Just as Hochmuth et al. (2025) find, the quantitative result strongly depends on the calibration of this parameter.

EoS energy and non-energy factors, ν A higher elasticity of substitution would imply more substitutability between energy and the capital-labor input in the production of the final consumption good. In this case, an increase in energy prices is expected to have a less severe negative effect on GDP and wages. I reanalyze GDP and wage decreases when adjusting ν from 0.2 to 0.4 while keeping the carbon tax constant. This allows for a clean comparison of the robustness of the economic distortions caused by the tax with respect to the EoS between energy and other inputs. The results, shown in online appendix E, figure 8, stay the same qualitatively and quantitatively.

Down-payment constraint, λ^{LTV} The importance of the financing frictions is determined by the down payment constraint. In the current calibration, a 10% down-payment is required. I check robustness of the results by relaxing the down-payment constraint, that is if everyone is allowed to borrow the full amount against the market interest rate when electrifying, implying a risk premium of 0%.

I show in figure 9 in online appendix E, that households are not affected by the financing conditions in their switching decisions. This can be explained by the fact that borrowing is still expensive, as interest rates rise sharply over the transition (figure 5, panel (c)). The high cost of borrowing makes that alleviating risk premia or increasing the space of borrowing does not affect borrowing at the margin. It does make borrowing for households who are already taking out credit cheaper. This means that the effects for the lower-end of the income distribution who do not electrify are not driven by arbitrary assumptions on the down-payment constraint.

Initial price premium, $A_{d_G,0}$ In the calibration I discuss the noisiness in determining the price premium of the green durable. I therefore check what happens if I lower the price premium from 2.55 to 2.2. Following, Kuhn & Schlattmann (2024), it is more likely to be lower than higher than 2.55. I show the welfare results remain qualitatively the same. Quantitatively, two things change slightly. First, the pain from investment lowers as the price of the green durable lowers. Hence the aggregate welfare loss from the tax in case of wasteful government spending decreases from -0.0696 to -0.0678. Second, the age group with the largest tax burden shifts to older ages. This is because the point of indifference between a drop in lifetime utility from durable investment and energy expenditures happens at a later age. Whereas first the main burden concentrated around ages 50 to 55, in this scenario it would be 53 to 58.

6 Front-loading the carbon tax

Front-loading the carbon tax offers the advantage that growth in the cumulative carbon stock slows earlier, and therefore a lower tax rate can achieve the same climate target. I find the endogenous tax rate when the tax is implemented in $T_{FL} = 15$ time periods for each instrument, like in the above setting. In figure 10, I plot the results of the endogenous tax rates when front-loading.

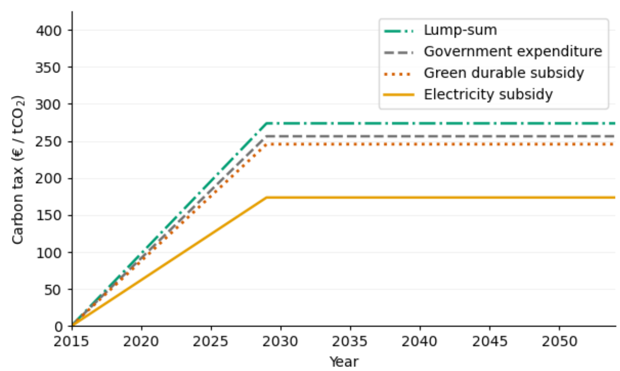


Figure 10: **Carbon tax front-loaded** path of carbon taxes per revenue-recycling instrument to meet the exogenous temperature target with front-loading

The qualitative order of tax rates between the instruments remains the same, however compared with the baseline case within instruments, all tax rates are significantly lower in the long run. Therefore, in the new steady state after the transition is complete, GDP and wages are higher than in the slow implementation case. However, because of front-loading, in the immediate short run (years 2015 - 2020) the carbon tax is higher, and since the productivity gains in the electricity sector have not yet fully materialized in the short run (up to $t = 15$), there is a large short-run GDP and wage dip in the front-loading case. This larger short-run dip, combined with a higher fossil fuel price in near-term periods, produce a larger negative aggregate welfare burden for generations alive when the policy is introduced. I visualize this timing trade-off for each instrument in figure 11.

The left panel shows, for each instrument, the GDP decline in the final steady state (y-axis) and the aggregate current welfare loss for alive generations (x-axis) for the baseline/slow-implementation (BL) and front-loaded (FL) case. The right-hand panel shows the equivalent comparison but with the welfare loss of the lowest income decile rather than the aggregate. A downward-sloping curve between FL and BL implies a timing trade-off: increasing the speed of phasing in raises GDP, but increases (decreases) the current welfare loss (gain). The solid lines show the trade-offs for the case with the electrification margin; the dashed lines show the results when the electrification margin is excluded.

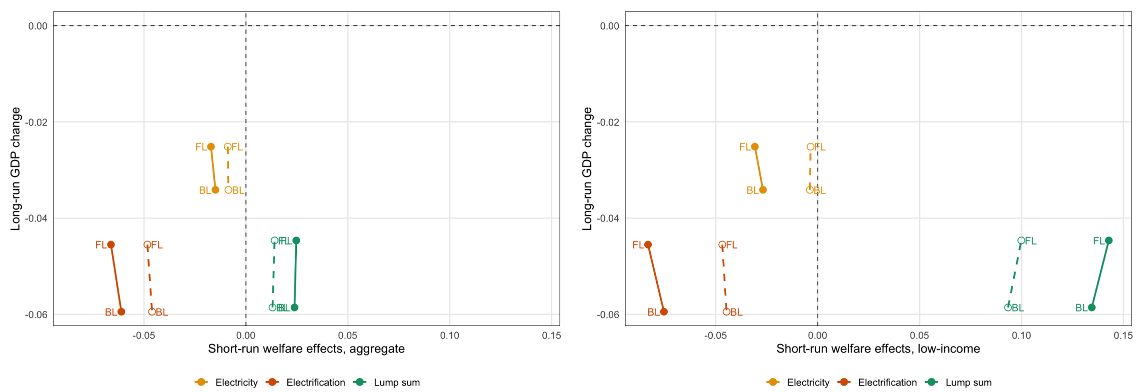


Figure 11: **Timing trade-off** GDP decline in the long run versus aggregate welfare change now (left) and low-income group welfare change now (right) for the baseline (BL) and front-loaded tax (FL) case. Solid line shows results with electrification margin, dashed line shows results without electrification margin. Government expenditure omitted.

The figure highlights several results. First, for all instruments, the long run GDP decrease relative to no policy is smaller when front-loading the tax, because of the lower tax rate (figure 10 versus figure 4). Second, a pattern already established above is clearly visible in this figure: the electrification margin affects welfare negatively after a carbon tax is introduced (excl. lump sum recycling), and this effect is significantly larger for low-income households. Third, comparing welfare outcomes between front- and back-loading the tax, front-loading increases the magnitude of the welfare effect: for subsidy recycling, welfare becomes more negative, while with lump-sum, it becomes more positive. These welfare effects are larger because of the electrification margin, especially for the poorest households. This implies that front-loading hurts low-income households the most, and the electrification margin is the cause. Because low-income households cannot adopt green technologies early, the higher short-run tax rate hurts them more. Lump-sum redistribution alleviates the timing trade-off.

7 Conclusion

In this paper, I study the effects of a carbon tax on efficiency and inequality in an economy containing both sides of the energy transition: electrification and decarbonization of electricity. I examine these effects under multiple revenue recycling instruments. I show that while carbon taxation robustly accelerates emission reduction, its effectiveness and welfare effects depend heavily on the revenue-recycling instrument. I also show that welfare effects are unequally distributed, largely because of the electrification margin.

I show that the electrification margin causes a large welfare loss at the lower-end of the income distribution, especially for middle-aged households via higher energy and durable expenditures. When revenues are recycled through a green-electricity subsidy, these effects are driving the welfare outcomes. On the other hand, they are overturned when households receive all tax revenues back in the form of a uniform lump-sum transfer. However, the latter comes at a cost of a lower GDP in the final steady state, caused by a higher maximum tax rate. Combining front-loading with lump-sum redistribution can offset this cost. Under this combination, front-loading does not decrease aggregate welfare, is more progressive, and mutes GDP losses.

When policy-makers are intent on reducing emissions, they have to take into account the horizontal and vertical inequalities these policies might create. Among the instruments considered in this paper and conditional on achieving the same climate target, lump-sum recycling protects exposed households but entails larger output losses. However, the optimal tax schedule and revenue distribution balancing both objectives remains an open question.

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Appendix

A Calibration data

For wealth and income statistics, I use the DNB household survey which consists of longitudinal data following about 2000 households. The survey contains multiple topics, such as work, pensions, mortgages, income, assets, liabilities and others. I use information about income, assets and liabilities from the AGI (Aggregated data on income) and AGW (Aggregated data on wealth) datasets. These datasets contain information about for example total assets or income, as a sum of individual components that were asked in the surveys. The data I use is from the years 2010 to 2020.

To estimate the income process, I constrain the sample to the ages 25-65. I delete observations with a negative net labor income or net labor income that exceeds 300.000 euros per year, to not include outliers.
For the wealth moments, I treat negative wealth levels as zero-wealth in order to match the models no unsecured borrowing constraint. Net worth is calculated as the sum of checking account, savings, bond holdings, stock holdings and mutual funds, cars and housing. As I am interested in homeowners and car owners making the investment into a cleaner option, for the wealth moments I only use data on homeowners. If I were to include renters, the sum of wealth would be lower and I would underestimate the financial ability of homeowners to make a green investment.

A.1 Heat pump uptake

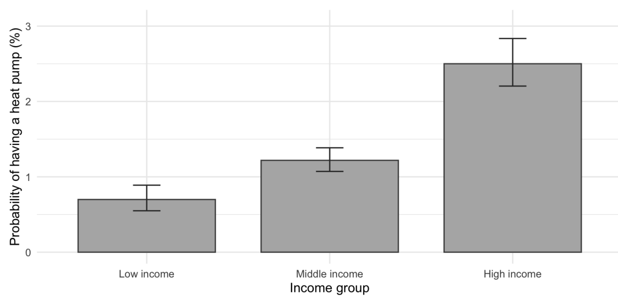


Figure 12: The predicted share of households for different income groups with a heatpump in 2018 using the WOon-dataset from Statistics Netherlands, for homeowners (37750 obs.), mean is 1.41% (dashed line). The correlation persists when controlling for square meters of the house and household age. The correlation is also found for 2021 and 2024.

B Climate Model

B.1 The Climate Model

The climate module is exactly taken from Douenne et al. (2025b). Temperature is given as

$$Z_{t+1} = Z_t + \epsilon \left(\zeta \mathcal{E}_t - Z_t \right), \tag{39}$$

where cumulative emissions evolve according to $\mathcal{E}_{t+1} = \mathcal{E}_t + E_t^M + E_t^{\text{ex}}$. I set Z_0, ϵ, ζ, $\mathcal{E}_0$, and the exogenous emissions $\{E_t^{\text{ex}}\}_{t \geq 0}$ externally, and take values from Douenne et al. (2025b).

B.2 Mapping model outcomes to emissions

In the model, I explicitly track fossil fuel demand, FF_t, as the sum of direct household and firm fossil fuel demand and indirect fossil fuel demand in electricity production:

$$FF_t = FF_{E,t} + FF_{C,t} + FF_{HH,t}. \tag{40}$$

The exogenous price of fossil fuels is calibrated as the price of 1000 kWh of energy, where 67% stems from gasoline and 33% from natural gas. The 670 kWh gasoline component corresponds to $\frac{670}{8.9} = 75.3$ liters of gasoline, and the 330 kWh natural gas component corresponds to $\frac{330}{10} = 33$ m^3 of natural gas. Using emission factors of 2.3 kg CO_2 per liter of gasoline and 1.9 kg CO_2 per m^3 of natural gas, one unit of fossil fuel in the model produces $75.3 \cdot 2.3 + 33 \cdot 1.9 = 235.85 kgCO_2$. Hence, model emissions implied by fossil fuel demand are given by $E_t^M = 235.85 \cdot FF_t$, with E_t^M measured in kg CO_2.

B.3 Temperature outcomes

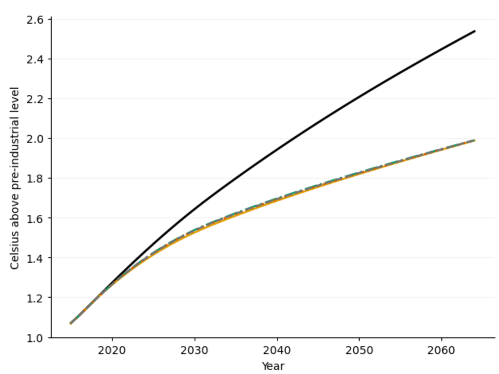


Figure 13: Temperature paths by revenue-recycling instrument in order to meet the exogenous temperature target (2 degrees Celsius)

C Lifetime Consumption Decomposition

At age j and time t, the household's budget constraint is:

$$\begin{aligned} &c_{i,j,t} + P^{\phi_{i,j,t}} e_{i,j,t} + a'_{i,j,t} + P_t^{d_{\phi_{i,j,t}}} \delta_{\phi_{i,j,t},t} = \\ &\quad \mathcal{T}_t + w_t y_{i,j} + (1+r_t) a_{i,j,t} - \pi(j,t) m_{i,j,t} + (\phi'_{i,j,t} - \phi_{i,j,t}) \left[m'_{i,j,t} - (1 - s_{d_G,t}) P_t^{d_G} + \chi P^{d_B} \right] \end{aligned} \tag{41}$$

where the left-hand side is consumption, energy expenditure, and savings, and the right-hand side contains transfers, labor income, asset income, maintenance costs, loan repayments, and the net cost of switching to a green durable. For compact notation, define:

$$EE_{i,j,t} \equiv P_t^{\phi_{i,j,t}} e_{i,j,t},$$
$$DE_{i,j,t} \equiv P_t^{d_{\phi_{i,j,t}}} \delta_{\phi_{i,j,t},t} + \pi(j,t)\, m_{i,j,t} - \left(\phi'_{i,j,t} - \phi_{i,j,t}\right)\left(m'_{i,j,t} - (1 - s_{d_G,t}) P_t^{d_G} + \chi P^{d_B}\right).$$

as energy expenditure and net durable expenditure respectively.

Next, consider a household i of age $j_{0,i}$ at the time the policy is introduced, $t = 0$. This household has $S_i \equiv J - 1 - j_{0,i}$ remaining periods of life. The constraints in (41) hold in all periods and are denominated in the consumption good of their own date, so they cannot be added directly. I therefore multiply the date-s constraint by Q_s, which converts each period's resources into units of consumption at $s = 0$, before summing over $s = 0, 1, \ldots, S_i$. This gives

$$E_0 \sum_{s=0}^{S_i} Q_s\, c_{i,j_{0,i}+s,\,s} =$$
$$E_0 \sum_{s=0}^{S_i} Q_s \left[\mathcal{T}_s + w_s y_{i,j_{0,i}+s} + (1 + r_s) a_{i,j_{0,i}+s,\,s} - a'_{i,j_{0,i}+s,\,s} - \mathrm{EE}_{i,j_{0,i}+s,\,s} - \mathrm{DE}_{i,j_{0,i}+s,\,s}\right]$$

Using the fact that $a'_{i,j,t} = a_{i,j+1,\,t+1}$, the discounted asset terms form a telescoping sum. Writing them out explicitly:

$$E_0 \sum_{s=0}^{S_i} Q_s \left[(1 + r_s) a_{i,j_{0,i}+s,\,s} - a'_{i,j_{0,i}+s,\,s}\right] = E_0 \Big[Q_0 \big[(1 + r_0) a_{i,j_{0,i},0} - a_{i,j_{0,i}+1,\,1}\big]$$
$$+ Q_1 \big[(1 + r_1) a_{i,j_{0,i}+1,\,1} - a_{i,j_{0,i}+2,\,2}\big] + \cdots + Q_{S_i} \big[(1 + r_{S_i}) a_{i,j_{0,i}+S_i,\,S_i} - a_{i,j_{0,i}+S_i+1,\,S_i+1}\big] \Big]$$

By construction $Q_s(1 + r_s) = Q_{s-1}$ for every $s \geq 1$. Hence, for each $s \geq 1$, the asset $a_{i,j_{0,i}+s,\,s}$ appears as $-Q_{s-1} a_{i,j_{0,i}+s,\,s}$ from the previous row and as $+Q_s(1 + r_s) a_{i,j_{0,i}+s,\,s} = +Q_{s-1} a_{i,j_{0,i}+s,\,s}$ from its own row, so the two cancel exactly: discounting at the market rate values a unit saved at $s - 1$ and its gross return at s identically in date-0 goods, and the intermediate asset positions therefore drop out of the identity. The final term $Q_{S_i}\, a_{i,j_{0,i}+S_i+1,\,S_i+1} = Q_{S_i}\, a_{i,J,T}$ is the discounted stock of assets left at death, which I denote as the financial bequest $a_{i,J}$. The telescoping sum therefore equals:

$$E_0 \sum_{s=0}^{S_i} Q_s \left[(1+r_s) a_{i,j_{0,i}+s,\,s} - a'_{i,j_{0,i}+s,\,s} \right] = (1+r_0)\, a_{i,j_{0,i},0} - E_0 \big[Q_{S_i}\, a_{i,J} \big] \tag{42}$$

Substituting this back into the original equation and moving the bequest $a_{i,J}$ to the left-hand side:

$$\underbrace{E_0 \sum_{s=0}^{S_i} Q_s\, c_{i,j_{0,i}+s,\,s} + Q_{S_i}\, a_{i,J}}_{\text{Lifetime Consumption}} = E_0 \Bigg[\underbrace{(1+r_0)\, a_{i,j_{0,i},0}}_{\text{initial wealth}} + \underbrace{\sum_{s=0}^{S_i} Q_s\, w_s\, y_{i,j_{0,i}+s}}_{\text{labor income}} + \underbrace{\sum_{s=0}^{S_i} Q_s\, \mathcal{T}_s}_{\text{transfers}} - \underbrace{\sum_{s=0}^{S_i} Q_s\, \mathrm{EE}_{i,j_{0,i}+s,\,s}}_{\text{energy exp.}} - \underbrace{\sum_{s=0}^{S_i} Q_s\, \mathrm{DE}_{i,j_{0,i}+s,\,s}}_{\text{durable exp.}} \Bigg] \tag{43}$$

This identity states that total lifetime consumption (including financial bequests) equals initial wealth plus the present value of all income flows, net of energy and durable expenditures, with every term expressed in units of consumption at $s = 0$.

Taking the difference $\Delta(\cdot)$ between the policy scenario and the no-policy baseline, and noting that initial wealth cancels:

$$\Delta E_0 \left[\sum_{s=0}^{S_i} Q_s\, c_{i,j_{0,i}+s,\,s} + Q_{S_i}\, a_{i,J} \right] = \Delta \sum_{s=0}^{S_i} Q_s\, \mathcal{T}_s + \Delta E_0 \sum_{s=0}^{S_i} Q_s\, w_s\, y_{i,j_{0,i}+s} - \Delta E_0 \sum_{s=0}^{S_i} Q_s\, \mathrm{EE}_{i,j_{0,i}+s,\,s} - \Delta E_0 \sum_{s=0}^{S_i} Q_s\, \mathrm{DE}_{i,j_{0,i}+s,\,s} \tag{44}$$

Online Appendix

A Solving the household problem

The value function is given by

$$V_j(a,\phi,m,y;P,\Omega)=\max\{V_j^{NA}(a,\phi,m,y;P,\Omega),V_j^{Electrify}(a,\phi,m,y;P,\Omega)+\mathcal{E}_{adj}\} \tag{1}$$

The preference shock μ_{Adj} causes that this extensive margin becomes a switching probability in the expectation, i.e. before realization of the shock. The probability is given by

$$p^{adj}(\mathbf{x};P,S)=1-\exp\left(-\exp\left(\frac{V_j^{Electrify}(\mathbf{x};P,\Omega)+\mu_{adj}-V_j^{NA}(\mathbf{x};P,\Omega)}{\sigma_{adj}}\right)\right) \tag{2}$$

such that

$$V^{\text{num}}=p^{\text{adj}}V^{\text{Electrify}}+\left(1-p^{\text{adj}}\right)V^{NA}, \tag{3}$$

$$q^{\text{num}}=p^{\text{adj}}q^{\text{Electrify}}+\left(1-p^{\text{adj}}\right)q^{NA}. \tag{4}$$

I individually solve for the intermediate functions. First for the last living period given the bequest value, then I use backward induction to solve for the remainder of the periods.
I start by solving the non-adjuster problem. I first solve for the static problem analytically. The dynamic problem then becomes

$$V_j^{NA}(a,\phi,m,y;P,S)=\max_{a'}\tilde{u}(\tilde{c})+\beta\mathbb{E}[V_{j+1}^{num}(a',\phi',m',y';P',S')] \tag{5}$$

subject to

$$\tilde{w}=wy+(1+r)a+\mathcal{T}-\delta_\phi P^{d_\phi}-\pi(j,t)m-a' \tag{6}$$

$$e=(1-\gamma)\frac{\tilde{w}}{p_\phi}+\gamma\frac{\bar{s}}{A_{s,\phi}},c=\gamma\left(\tilde{w}-\frac{p_\phi}{A_{s,\phi}}\bar{s}\right) \tag{7}$$

$$\tilde{u}(\tilde{c})=\frac{\tilde{c}^{1-\sigma}}{1-\sigma}\text{ with }\tilde{c}=c^\gamma\left(A_{s,\phi}e-\bar{s}\right)^{1-\gamma} \tag{8}$$

$$m'=(1+r+r^m-\pi(j,t))\,m,\quad \phi'=\phi a'\geq 0 \tag{9}$$

where $A_{s,\phi}\equiv\phi A_{s,G}+(1-\phi)A_{s,B}$ and $p_\phi\equiv\phi P^E+(1-\phi)(P^{FF}+\tau_c)$. I solve this problem using EGM (where q^{num} is the marginal utility of expenditure) with upper envelope as described in Druedahl (2021).

Given value function $V_j^{NA}(\mathbf{x}; P, S)$ and policy functions $a_j'(\mathbf{x}; P, S)$, $c_j(\mathbf{x}; P, S)$, $e_j(\mathbf{x}; P, S)$, I continue by solving for the electrifier value function, by recognizing that this is an interpolation of the non-adjuster problem

$$V_j^{Electrify}(a, \phi, m, y; P, \Omega) = \max_{m'} V_j^{NA}(\tilde{a}, \phi' = 1, m', y; P, \Omega) \tag{10}$$

$$\tilde{a} = a - (P^G(1 - s_{dG}) - \chi P^B) + m', \quad m' \leq \lambda(1 - s_{d_G})P^G \tag{11}$$

Which gives policy function $\phi_j'(\mathbf{x}; P, S)$,$m_j'(\mathbf{x}; P, S)$ and value function $V_j^{Electrify}(\mathbf{x}; P, S)$

B Solving the Production side

B.1 Output-good-producing firm

A representative firm produces output goods which can be used for production or converted in durable goods. Inputs are capital, labor and electricity. The technology is a constant returns to scale CES

$$Y^C(K_C, L_C, E_C, FF_C) = \left[(1-\psi)\left(A_{KC}K_C^{\alpha_c}L_C^{1-\alpha_c}\right)^{\frac{\nu-1}{\nu}} + \psi A_{MC} M^{\frac{\nu-1}{\nu}}\right]^{\frac{\nu}{\nu-1}} \tag{12}$$

Gives first order conditions

$$r + \delta = \left[(1-\psi)B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}}\right]^{\frac{1}{\nu-1}} (1-\psi)\, B^{\frac{\nu-1}{\nu}-1}\, A_{KC}\alpha_c\, K_C^{\alpha_c-1} L_C^{1-\alpha_c} \tag{13}$$

$$w = \left[(1-\psi)B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}}\right]^{\frac{1}{\nu-1}} (1-\psi)\, B^{\frac{\nu-1}{\nu}-1}\, A_{KC}(1-\alpha_c)\, K_C^{\alpha_c} L_C^{-\alpha_c} \tag{14}$$

$$P^E = \left[(1-\psi)B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}}\right]^{\frac{1}{\nu-1}} \psi\, M^{\frac{\nu-1}{\nu}-1} \left(M^{\frac{1}{\iota}} \kappa E_C^{\frac{\iota-1}{\iota}-1}\right) \tag{15}$$

$$P^{FF} + \tau_c = \left[(1-\psi)B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}}\right]^{\frac{1}{\nu-1}} \psi\, M^{\frac{\nu-1}{\nu}-1} \left(M^{\frac{1}{\iota}} (1-\kappa) FF_C^{\frac{\iota-1}{\iota}-1}\right) \tag{16}$$

where

$$B = A_{KC}K_C^{\alpha_c}L_C^{1-\alpha_c} \tag{17}$$

$$M = \left(\kappa E_C^{\frac{\iota-1}{\iota}} + (1-\kappa)FF_C^{\frac{\iota-1}{\iota}}\right)^{\frac{\iota}{\iota-1}}. \tag{18}$$

B.2 Durable good producers

In this economy, the durable goods can be produced by converting the consumption good (with cost 1, as its the numeraire) at some rate A_G, A_B, such that production functions are given as

$$Y^{D_G}(C_G) = A_G C_G \tag{19}$$

$$Y^{D_B}(C_B) = A_B C_B \tag{20}$$

These imply input demand functions

$$C_G = \frac{1}{A_G} Y^{D_G} \tag{21}$$

$$C_B = \frac{1}{A_B} Y^{D_B} \tag{22}$$

and prices $P^G = \frac{1}{A_G}$, $P^B = \frac{1}{A_B}$

B.3 Electricity firm

The profits of the electricity firm are given as in equation (25) in the paper. Equivalently, I can write

$$\Pi = P_E \big(\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}}\big)^{\frac{\lambda}{\lambda-1}} - P_{E_R} E_R - P_{E_F} E_F \tag{23}$$

subject to the technology constraints

$$E_R = A_R K_R \tag{24}$$

$$E_F = A_F [\zeta^{\frac{1}{\omega}} (A_{KF} K_F)^{\frac{\omega-1}{\omega}} + (1-\zeta)^{\frac{1}{\omega}} FF_E^{\frac{\omega-1}{\omega}}]^{\frac{\omega}{\omega-1}} \tag{25}$$

Which implies I can split up the problem in two parts. The first step to solving the electricity firm problem for the supply function, is to choose E_F, E_R to minimize cost subject to the constraint of producing some level of electricity $\bar{E}$. The Lagrangian reads

$$\mathcal{L} = P_{E_R} E_R + P_{E_F} E_F + \Lambda(\bar{E} - \big(\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}}\big)^{\frac{\lambda}{\lambda-1}})$$

The first order conditions are

$$\frac{d\mathcal{L}}{dE_R} \implies \Lambda \alpha_R^{\frac{1}{\lambda}} (E_R)^{\frac{\lambda-1}{\lambda}-1} \big(\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}}\big)^{\frac{\lambda}{\lambda-1}-1} = P_{E_R} \tag{26}$$

$$\frac{d\mathcal{L}}{dE_F} \implies \Lambda \alpha_F^{\frac{1}{\lambda}} (E_F)^{\frac{\lambda-1}{\lambda}-1} \big(\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}}\big)^{\frac{\lambda}{\lambda-1}-1} = P_{E_F} \tag{27}$$

$$\frac{d\mathcal{L}}{d\Lambda} \implies \bar{E} = \big(\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}}\big)^{\frac{\lambda}{\lambda-1}} \tag{28}$$

Dividing through

$$\frac{P_{E_R}}{P_{E_F}} = \frac{\alpha_R^{\frac{1}{\lambda}}(E_R)^{\frac{\lambda-1}{\lambda}-1}}{\alpha_F^{\frac{1}{\lambda}}(E_F)^{\frac{\lambda-1}{\lambda}-1}} \tag{29}$$

Solve for E_R and plug in

$$(\frac{P_{E_R}}{P_{E_F}})^{-\lambda}\frac{\alpha_R}{\alpha_F}E_F = E_R \implies \bar{E} = \left(\alpha_R^{\frac{1}{\lambda}}((\frac{P_{E_R}}{P_{E_F}})^{-\lambda}\frac{\alpha_R}{\alpha_F}E_F)^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}}E_F^{\frac{\lambda-1}{\lambda}}\right)^{\frac{\lambda}{\lambda-1}}$$

Rewriting:

$$\bar{E} = \left(\alpha_R^{\frac{1}{\lambda}}((\frac{P_{E_R}}{P_{E_F}})^{-\lambda}\frac{\alpha_R}{\alpha_F})^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}}\right)^{\frac{\lambda}{\lambda-1}}E_F \tag{30}$$

such that

$$E_F = \frac{\bar{E}}{\left(\alpha_R^{\frac{1}{\lambda}}((\frac{P_{E_R}}{P_{E_F}})^{-\lambda}\frac{\alpha_R}{\alpha_F})^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}}\right)^{\frac{\lambda}{\lambda-1}}} \tag{31}$$

where input demand scales linearly in the amount produced $\bar{E}$, taking all prices as given.

This implies that for any output $\bar{E}$, input demand is given by

$$E_F = \alpha_F\left(\frac{P_{E_F}}{P_E}\right)^{-\lambda}\bar{E} \tag{32}$$

$$E_R = \alpha_R\left(\frac{P_{E_R}}{P_E}\right)^{-\lambda}\bar{E} \tag{33}$$

and the natural price index for E is

$$P_E = (\alpha_R P_{E_R}^{1-\lambda} + \alpha_F P_{E_F}^{1-\lambda})^{\frac{1}{1-\lambda}} \tag{34}$$

Similarly, for the intermediate electricity from renewable and fossil, from the first order conditions, prices for these are given by marginal costs

$$P_{E_R} = \frac{r+\delta}{A_R} \tag{35}$$

$$P_{E_F} = \frac{1}{A_F}(\zeta((r+\delta)/A_{KF})^{1-\omega} + (1-\zeta)(P_{FF}+\tau_c)^{1-\omega})^{\frac{1}{1-\omega}} \tag{36}$$

and input demands are

$$K_R = \frac{1}{A_R} E_R \tag{37}$$

$$K_F = \zeta \left(\frac{r+\delta}{A_{KF} P_{E_F}} \right)^{-\omega} E_F \tag{38}$$

$$FF_E = (1-\zeta) \left(\frac{P_{FF} + \tau_c}{P_{E_F}} \right)^{-\omega} E_F \tag{39}$$

C Equilibrium algorithm

C.1 No tax, stationary equilibrium

1. **Initialization.**
   (a) Fix a tolerance level $\varepsilon_r > 0$ and a damping parameter $\eta_r \in (0,1)$.
   (b) Choose an initial guess for the interest rate $r^{(0)}$.

2. **Outer iteration** $i = 0, 1, 2, \ldots$ (until convergence):
   (a) Given $r^{(i)}$, compute the implied price of electricity and the prices of durables as marginal costs
   (b) Using a non-linear solver, given $r^{(i)}$ and P^E, solve for (K_C, E_C, FF_C, w) such that the labor market clears and the firm's first-order conditions are satisfied, described in appendix B.1.
   (c) Given the price vector $P = (P^{FF}, P^G, P^B, P^E, r^{(i)}, w)$, compute the value functions $V_j^{NA}(a, \phi, m, y; P, S)$, $\quad V_j^A(a, \phi, m, y; P, S)$ and policy functions $\phi_j'(a, \phi, m, y; P, S)$, $m_j'(a, \phi, m, y; P, S)$, $a_j'(a, \phi, m, y; P, S)$, $c_j(a, \phi, m, y; P, S)$, $e_j(a, \phi, m, y; P, S)$.
   (d) Compute the stationary distribution μ such that the bequest conditions hold
   (e) From the distribution, compute household aggregate demand for each good.
   (f) Use market clearing for electricity to get sectoral output $Y^E = E_{hh} + E_C$ and input demands plus intermediate-input prices as described in appendix B.3.
   (g) Check capital market clearing,

$$A = K_C + K_{E_R} + K_{E_F}, \tag{40}$$

and the aggregate resource constraint (using the input-demands from appendix B.2 to get C_B, C_G).

$$Y = C_{hh} + C_B + C_G + I_{E_R} + I_{E_F} + I_C \\ + P^{FF}(FF_{hh} + FF_C + F_E) + \int \phi\pi(j)M - (\phi' - \phi)M' \, d\mu. \quad (41)$$

Let $r^{(i')}$ denote the interest rate that clears the capital market given the aggregates computed above. If $|r^{(i')} - r^{(i)}| < \varepsilon_r$ and the resource constraint holds, stop. Otherwise update

$$r^{(i+1)} = \eta_r \, r^{(i)} + (1 - \eta_r) \, r^{(i')}$$

and return to step 2(a).

C.2 Transition

1. **Initialization.**

   (a) Fix a time horizon $t = 0, \ldots, T$, tolerance levels $\varepsilon_r > 0$, and a damping parameter $\eta_r \in (0, 1)$.

   (b) Choose an initial guess for the interest-rate path $\{r_t^{(i=0)}\}_{t=0}^T$ and for the policy redistribution instrument.

   (c) Given the path for productivity $\{A_{G,t}, A_{KR,t}\}_{t=0}^T$ and taxes $\{\tau_{c,t}\}_{t=0}^T$, calculate the implied sequences for $\{P_t^E\}_{t=0}^T$ and $\{w_t\}_{t=0}^T$ from the firm problem.

2. **Outer iteration** $i = 0, 1, 2, \ldots$ (until convergence):

   (a) Compute the value function of the stationary equilibrium in the final period T, $V_j^T(a, \phi; Y, P, \mathcal{T})$. By backward induction, taking the price sequences as given, compute the value function $V_j^t(a, \phi; Y, P, \mathcal{T})$ for $t = T - 1, ..., 0$. Using the policy functions, simulate the distribution forward for T periods starting from the initial distribution μ_0, being the initial cross-sectional distribution of households μ_0 as the steady state at $t = 0$

   (b) From the distribution, for each time period t calculate aggregate capital holdings $\int a(r_i) d\mu_t$ and aggregate electricity demand from households $E_{hh} = \int \phi e d\mu_t$

   (c) **Inner iteration,** $j = 0, 1, 2, \ldots$ (until convergence): solve the static production problem to clear the market. For each time period t:

i. Guess an interest rate r_j, calculate $P_j^E(r_j)$.

ii. Given r_j, $P_j^E(r_j)$ calculate K_C and E_C from the first order conditions of the final consumption good firm.

iii. Given E_{hh}, E_C and r_j calculate K^{E_R} and K_{E_F}

iv. Check if $\int a(r_i) d\mu_t = K_C(r_j) + K^{E_R}(r_j) + K_{E_F}(r_j)$. If not, update r_j and go back to ii.

(d) Check if $\{r_t^{(i)}\}_{t=0}^T = \{r_t^{(j)}\}_{t=0}^T$, also for the instrument, else update

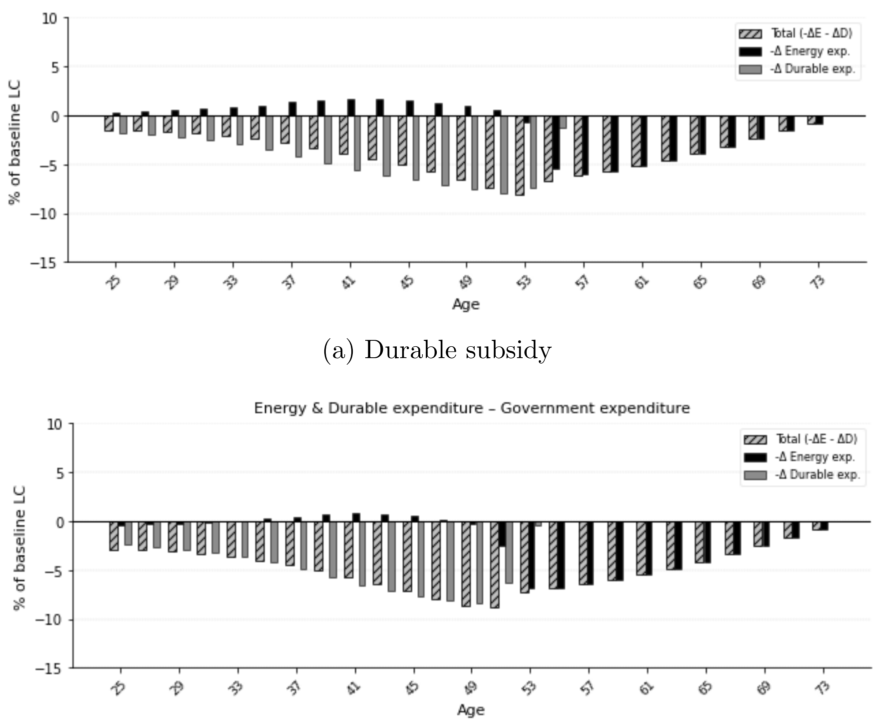
$$\{r_t^{(i+1)}\}_{t=0}^T = \eta_r \{r_t^{(i)}\}_{t=0}^T + (1-\eta_r)\{r_t^{(j)}\}_{t=0}^T$$

as well as for the instrument at hand, and continue with step 1.c.

D Extra Results

D.1 Heterogeneous expenditure changes

Figure 10 for durable subsidy and government expenditure

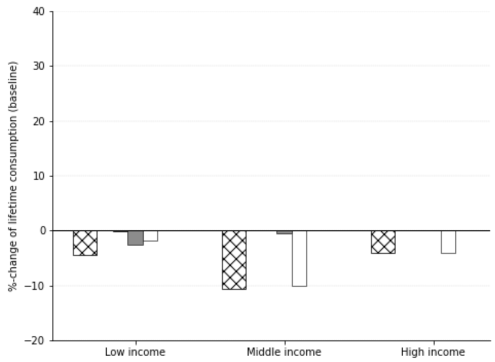


(a) Durable subsidy

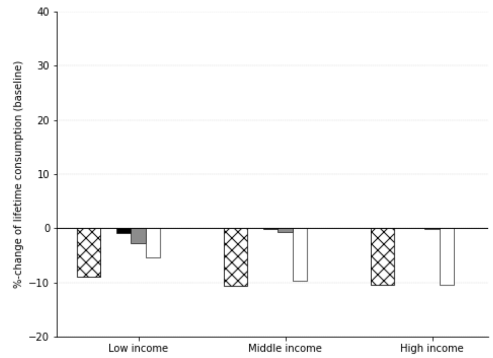
(b) Government expenditure

Figure 1: Heterogeneous expenditure responses across the age space for $z = 0$.

D.2 Consumption decomposition



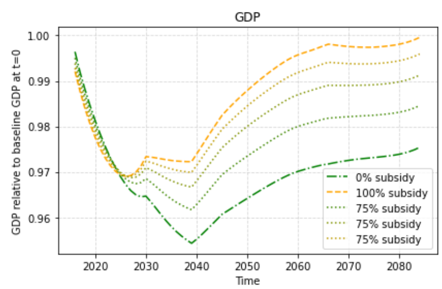
(a) Durable subsidy



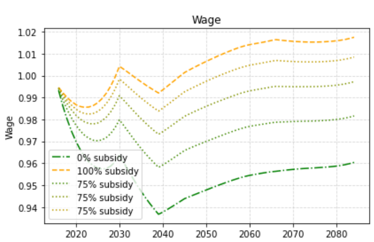
(b) Government expenditure

Figure 2: Heterogeneous expenditure responses across the age space for $z = 0$.

D.3 GDP and prices for combinations of lump-sum and electricity subsidy

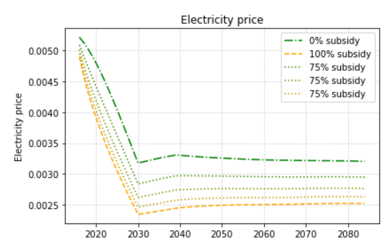


(a) GDP



(b) Wage

(c) Interest rate



(d) Electricity price

Figure 3: GDP (top) and prices (bottom) over the course of the transition for both subsidies and a combination of the two subsidies

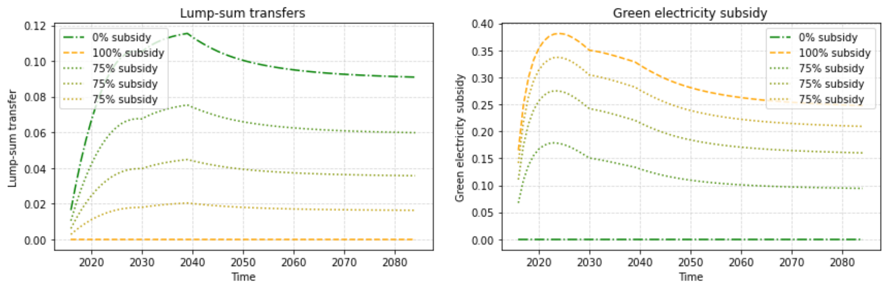
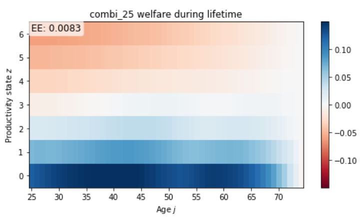
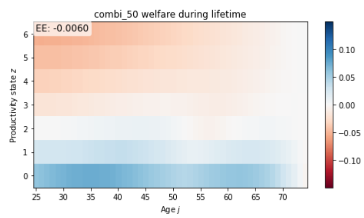


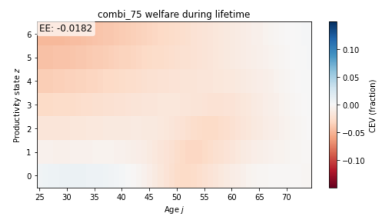
Figure 4: Policies. Right: Lump-sum, left: subsidy



(a) 25% to subsidies



(b) 50% to subsidies

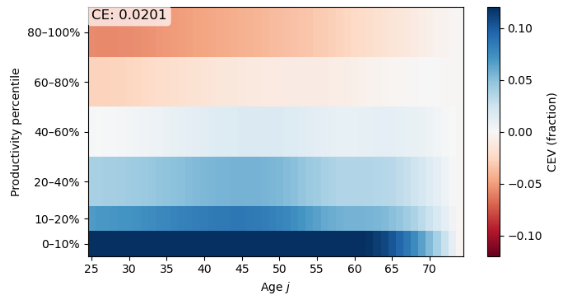


(c) 75% to subsidies

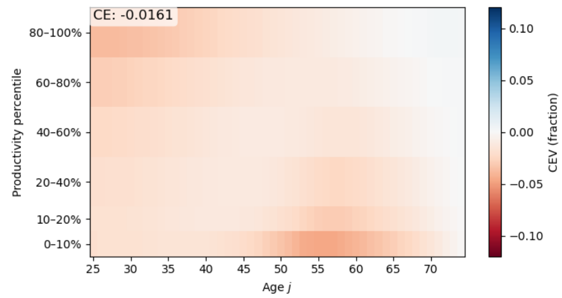
Figure 5: Consumption Equivalence different policy combinations

E Robustness

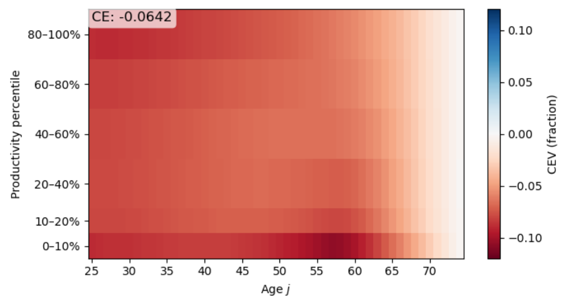
1. Idiosyncratic income risk



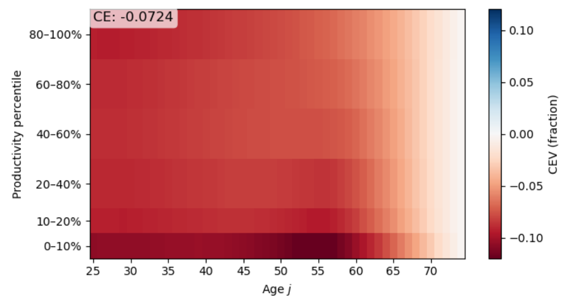
(a) Lump-sum



(b) Electricity subsidy



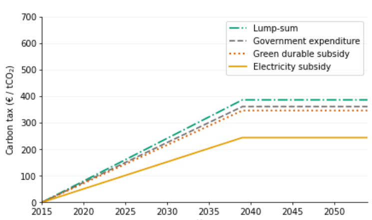
(c) Electrification subsidy



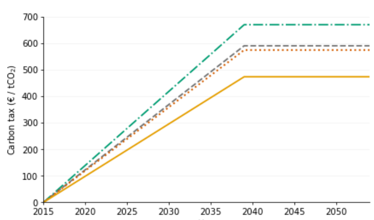
(d) Government spending

Figure 6: **Robustness check 1** Replication of Consumption Equivalence without income risk

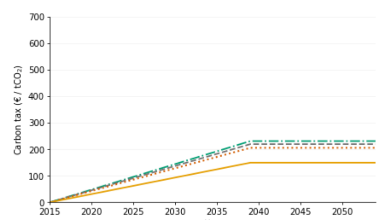
2. EoS renewable and fossil electricity, λ



(a) Baseline ($\lambda = 1.8$)



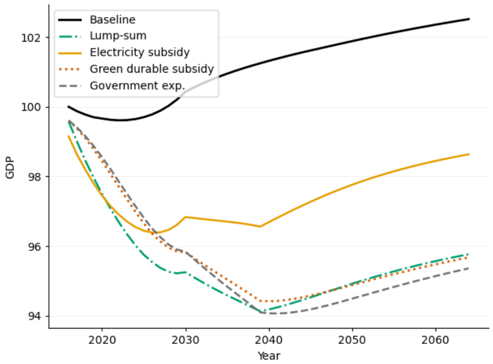
(b) Low elasticity ($\lambda = 0.5$)



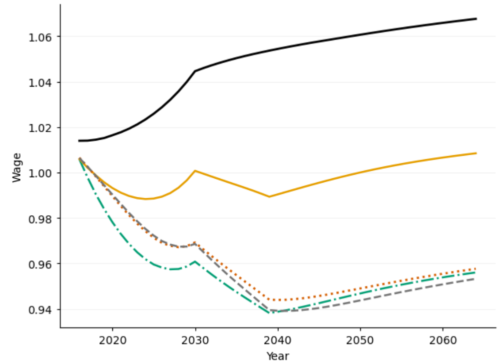
(c) High elasticity ($\lambda = 5$)

Figure 7: **Robustness check 2** $\tau_{c,t}$ for several EoS renewable and fossil electricity, λ

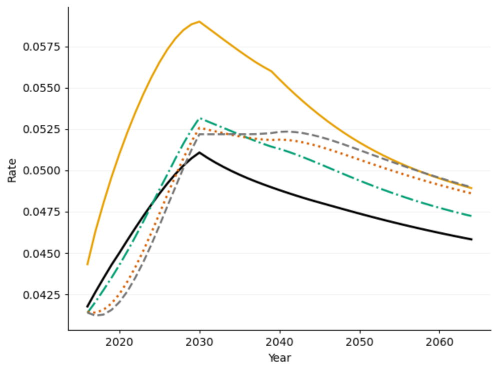
3. EoS energy and non-energy factors, ν to 0.4



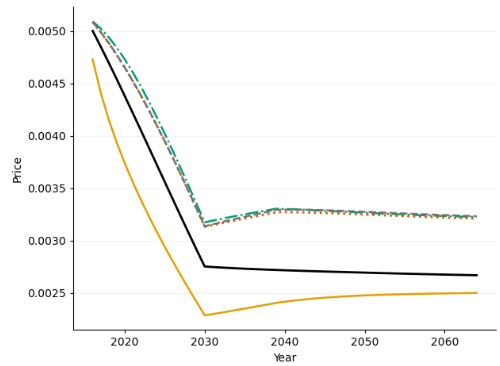
(a) GDP



(b) Wage



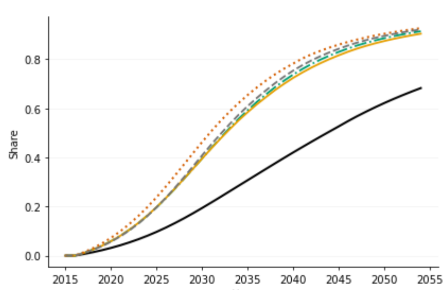
(c) Interest rate



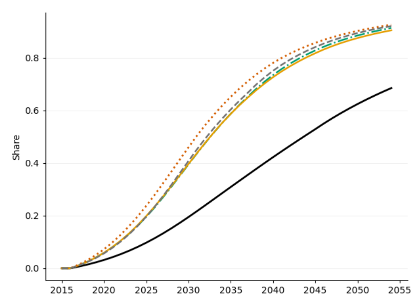
(d) Electricity price

Figure 8: **Robustness check 3** GDP and prices with elasticity of substitution between energy and other inputs, ν, equal to 0.4.

4. Down-payment constraint, λ^{LTV}



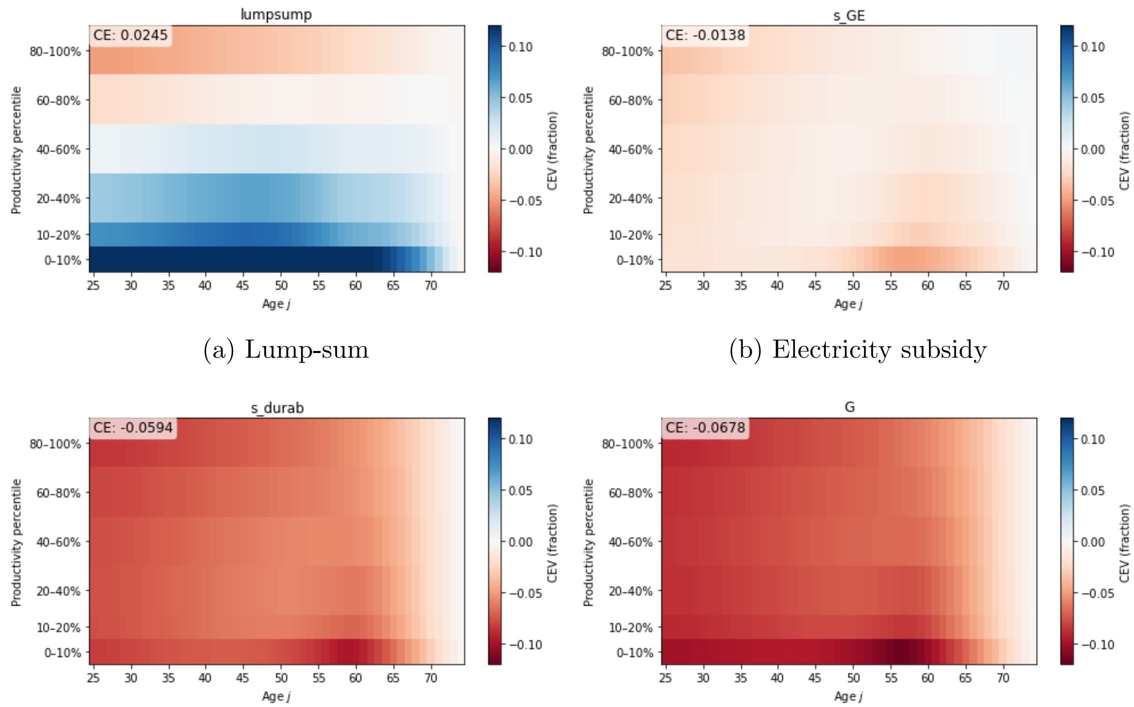
(a) Down-payment constraint



(b) No borrowing constraints

Figure 9: **Robustness check 4** Household electrification behavior under different calibrations of the down-payment constraint.

5. Initial price premium, $A_{d_G,0}$ to 2.2



(a) Lump-sum

(b) Electricity subsidy

(c) Electrification subsidy

(d) Government spending

Figure 10: **Robustness check 5** Replication of consumption equivalence with 2.2 price premium green durable

References

Druedahl, J. (2021). A guide on solving non-convex consumption-saving models. *Computational Economics*, *58*(3), 747–775.